

Dark Pool Trading and Information Acquisition

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Theory suggests that dark pools may facilitate or discourage information acquisition. We find that more dark pool trading leads to greater information acquisition. We measure information acquisition using stock price dynamics around earnings announcements. To overcome endogeneity concerns, we exploit a large exogenous decrease to dark pool trading that results from the implementation of the Security and Exchange Commission's (SEC's) Tick Size Pilot Program. The results cannot be explained by lit venue liquidity, algorithmic trading, or informational efficiency. A battery of additional tests, such as documenting a shift in SEC EDGAR searches, supports the information acquisition interpretation. (*JEL* G14, G18)

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Among the most dramatic changes in financial markets in the last few decades is the rise of dark venues. Dark venues, or dark pools, are equity trading venues in which traders buy and sell stocks without publicly displaying their orders. The market share of dark pools in the United States grew from 7.5% in 2008 to 12.6%

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in 2015 (Rosenblatt Securities 2021). The rapid growth has raised regulatory concerns about the impact of dark pool trading on price discovery in the financial markets.¹ Unlike trades made on stock exchanges, which directly affect prices, trades made on dark pools typically free-ride on the price discovery function of the exchanges, as they passively execute at prices derived from exchanges (Zhu 2014; Nimalendran and Ray 2014).

Price discovery is the process by which new information is incorporated into asset prices (O'Hara 2003). It consists of two components: the acquisition of new information, and the incorporation of existing information into asset prices (Brunnermeier 2005; Weller 2018). Both components can be referred to by various terms; for clarity, we refer to the former as “information acquisition” and the latter as “informational efficiency.”² Both components are important, as both can affect price discovery and therefore the allocation of risk and capital. While prior studies have examined the relationship between dark pool trading and informational efficiency and arrived at conflicting conclusions, the effect of dark pool trading on information acquisition has not been examined.³ Information acquisition is particular important to the allocation of risk and capital. The idea that prices are a useful source of information goes back to at least Hayek (1945). Prices aggregate diverse pieces of information produced by traders. Real decision-makers can use this information when making decisions or determining contractual performance (Bond, Edmans, and Goldstein 2012).⁴ The extent to which equity markets improve the efficiency

¹ In 2014, then-SEC-Chair Mary Jo White (2014) remarked that “we must continue to examine whether dark pool trading volume is approaching a level that risks seriously undermining the quality of price discovery provided by lit venues.”

² Brunnermeier (2005) distinguishes between two concepts: “informational efficiency” and “informativeness.” We use “informational efficiency” in the same sense that Brunnermeier (2005) does, and we use “information acquisition” in the same sense that Brunnermeier uses “informativeness.” Two additional terms in the literature, “market efficiency” and “price efficiency,” are considered synonyms for informational efficiency (Han, Tang, and Yang 2016).

³ For example, using data from 11 dark pools in the United States, Buti, Rindi, and Werner (2011) find that dark trading improves market quality measures and informational efficiency. Based on the implementation of the Tick Size Pilot Program, Albuquerque, Song, and Yao (2020) document that stocks subjected to the “trade-at” provision experience larger price errors and price delay, suggesting that dark trading improves informational efficiency on an intraday basis. In contrast, Hatheway, Kwan, and Zhen (2017) employ proprietary trade-by-trade data for 116 stocks stratified by market capitalization in the U.S. market and find the opposite. In addition to evidence on equity markets, Fleming and Nguyen (2013) conclude that dark trading improves price efficiency and overall price discovery of the U.S. Treasuries. A few recent studies also add international evidence to this literature. Using dark trading data from the Australian Securities Exchange, Comerton-Forde and Putniņš (2015) focus on the effect of dark trading on price efficiency and overall price discovery. Exploiting minimum price improvement regulation in Canada and Australia as shocks to dark trading, Foley and Putniņš (2016) document that different types of dark pools could have heterogeneous effects on market quality and informational efficiency.

⁴ Price informativeness in secondary markets affect the efficiency of real decisions in at least two important ways. First, as corporate managers’ contracts are typically tied to stock prices, the extent to which stock prices approximate fundamental values affects corporate managers’ incentives to exert costly effort to improve the firm’s fundamental value. Second, there is a feedback effect from market prices to real decisions. Prior research has found evidence that managers learn from the private information contained in stock prices to guide corporate decisions, such as corporate investments (e.g., Chen, Goldstein, and Jiang 2007), mergers and acquisitions (e.g., Betton et al. 2014), and corporate cash savings (e.g., Fresard 2010). Bond, Edmans, and Goldstein (2012) provide a detailed survey of both theoretical and empirical studies on the real effects of financial markets.

of real decisions depends on the new information produced by traders and conveyed to decision-makers. This paper examines whether dark pool trading facilitates or discourages information acquisition.

Theoretical arguments reason that dark pools may encourage or discourage information acquisition through their impact on trading costs. Grossman and Stiglitz (1980) show that for investors to acquire new information they must expect to profit from it. By hiding their orders in dark pools before execution, informed investors can enter and exit transactions at low cost (Boulatov and George 2013). Hu, Jones, and Zhang (2021), Reed, Samadi, and Sokobin (2020), Balakrishnan et al. (2020), and Ye and Zhu (2018) all find evidence that informed traders utilize dark pools. The lower transaction costs make lower-value information acquisition opportunities worth pursuing, which should increase information acquisition.

Alternatively, if dark pools result in a partial separating equilibrium whereby exchanges become relatively more attractive to informed traders by providing immediacy, then the informational efficiency component of price discovery may increase, but the cost of trading may also rise (Zhu 2014).⁵ As a result of the higher trading costs, investors may expend less resources to acquire and trade on information, which would decrease information acquisition (Grossman and Stiglitz 1980; Diamond and Verrecchia 1981; Verrecchia 1982; Kyle 1985, 1989). The direction of the effect is critical, because one cannot understand whether dark pools improve or harm price discovery—which is of first-order importance for the allocation of risk and capital—without understanding how dark pools affect information acquisition.

We focus on information acquisition occurring before quarterly earnings news because earnings are a well-documented, economically important, consistently quantifiable source of information carefully followed by market participants.⁶ We use newly available weekly dark pool trading data provided by the Financial Industry Regulatory Authority (FINRA). Since the week of May 11, 2014, FINRA has mandated that each dark pool report its weekly trading volume information in an effort to increase market transparency. Using the newly available FINRA data, we construct a sample of 40,150 firm-quarter observations from September 2014 to December 2017. We compute the level of dark pool trading as the ratio of the trading volume executed on dark pools to the consolidated trading volume.

⁵ In Zhu's (2014) model, investors face a trade-off between the transaction costs and urgency for trading when choosing to trade on-exchange or in dark pools. Dark pools offer lower transaction costs than exchanges, which is attractive for both informed and uninformed traders. Because in a typical dark pool, there is no market maker to absorb excess order flows, informed traders, who are more likely to trade in the same direction as each other, face a higher execution risk in dark pools relative to uninformed traders. Therefore, Zhu (2014) predicts that exchanges will be relatively attractive to informed traders, and dark pools will be relatively attractive to uninformed traders.

⁶ While we focus on the setting of quarterly earnings announcements, the mechanisms linking dark trading and information acquisition likely apply generally.

We use two measures of information acquisition. Both are meant to capture how well prices contain future fundamental (earnings) information about a firm. The first is Weller's (2018) jump ratio. The jump ratio divides the price adjustments during earnings announcements by the price adjustments during and before the earnings announcements. A lower ratio indicates more information acquisition. The idea behind it is straightforward: while the numerator captures the surprise element of the earnings announcement, the denominator captures the surprise element plus the earnings-related information acquisition before the earnings announcement.

The second measure of information acquisition is the future earnings response coefficient (FERC). The FERC is a longer horizon information acquisition measure. It measures the association between a firm's future earnings and a firm's current returns. A higher FERC means that more information about future earnings is acquired and revealed in current returns. The FERC measure and variations of it have been in the recent finance literature (Glosten, Nallareddy, and Zou 2020; Israeli, Lee, and Sridharan 2017; Bai, Philippon, and Savov 2016).

Before dealing with endogeneity issues, we perform a traditional ordinary least squares analysis of the level of dark pool trading on the two different measures of information acquisition. The benefit of this analysis is that it allows us to make statements about the broad universe of stocks; the limitation is that only an association can be established. We find that a higher level of dark pool trading is associated with more information acquisition (i.e., a smaller jump ratio and a higher FERC).

Endogeneity issues complicate any effort at determining the causal effects of dark pool trading. Unobservable factors that affect investors' decisions to trade on exchanges or in dark pools could also affect the market's response to earnings news related to the stock being traded. In addition, the level of information acquisition could drive the level of dark pool trading. To identify the causal effect of dark pool trading on information acquisition, we use the Security and Exchange Commission's (SEC's) Tick Size Pilot Program as a plausible source of exogenous variation in dark pool trading. Researchers have studied other effects of this program, which was enacted in October 2016 and implemented on a staggered basis over two years (Albuquerque, Song, and Yao 2020; Rindi and Werner 2017; Lee and Watts 2021; Li, Ye, and Zheng 2018; Farley, Kelley, and Puckett 2018; Chakrabarty, Cox, and Upson 2021; Comerton-Forde, Grégoire, and Zhong 2019). The upside of focusing on this program is that it alleviates endogeneity concerns; the downside is that the results are based on a subset of stocks and therefore may not hold universally.

The stocks in group 1 (G1) of the Tick Size Pilot Program must be quoted in 5-cent increments. The stocks in group 2 (G2) are subject to the requirements applying to G1 stocks and are also required to be traded in 5-cent increments or at the midpoint of the NBBO. The stocks in group 3 (G3) are subject to the requirements applying to G2 stocks and also to the "trade-at" provision, which

requires orders to be executed in lit venues unless dark venues can execute them at a meaningfully better price. By increasing the costs of trading in dark pools, the “trade-at” provision decreases trading volume in dark pools.

Our difference-in-differences (DiD) analysis includes all G2 and G3 stocks of the Tick Size Pilot Program. A comparison of effects between G2 and G3 holds constant the quoting and trading increments and thereby isolates any dark pool trading effects. We use G3 stocks as the treatment stocks and G2 stocks as the control stocks in the analysis. Note that the treatment group experiences a negative shock to its level of dark pool trading. We use a $[-120, 120]$ day window around the effective dates of the pilot program to conduct the DiD analysis. The calendar date of the window varies for each stock, as the Tick Size Pilot Program was launched on October 3, 2016, and implemented on a staggered basis over two years. The staggered introduction allows us to better control for the impact of unobserved variables, because time series changes unrelated to the Tick Size Pilot Program are unlikely to occur simultaneously with the treatment.

In the treatment group, the average market share of dark pools drops by 30% (from 14.6% to 10.3%) from the 20 weeks before to the 20 weeks after the implementation dates of the pilot program. The DiD analysis shows a smaller preemption of upcoming earnings news (higher jump ratio) and a smaller FERC for treated firms than for control firms. Both results are consistent with less dark pool trading decreasing information acquisition.⁷ The information acquisition interpretation is further supported through a battery of additional tests. In sum, the results are consistent with the idea that dark pools, with their lower transaction costs, are appealing to traders who are deciding whether to expend resources collecting information about upcoming earnings news. Immediacy may be a secondary concern in this setting as informed traders can trade well in advance of the earnings news release.

While the DiD approach can help overcome endogeneity problems, the implementation of the Tick Size Pilot Program may have led to other changes that drive our results, so it is important to explore alternative explanations. Given the research question and identification strategy, we acknowledge three possible confounding factors: lit venue liquidity, algorithmic trading, and informational efficiency. When we explore these alternative channels, we find that they do not explain our results.

This study contributes to two streams of literature. First, it adds to the emerging literature on the consequences of dark pool trading. Prior empirical research on dark pool trading has focused on either price discovery overall or the

⁷ In interpreting the results of this paper, one should note the context of the particular market environment. One cannot necessarily extrapolate the results to substantially different settings. For instance, the results would likely be different if examined in a setting where 90% of trading activity occurred in dark pools.

informational efficiency component of price discovery on an intraday basis.⁸ Information acquisition and informational efficiency need not be similarly affected by dark pool accessibility. For example, Hatheway, Kwan, and Zhen (2017) shows that less dark pool trading improves informational efficiency. One possible reason is that, as dark pools divert a large volume of trades from exchanges, the market depth on exchanges decrease, transitory price movements increase, and so the informational efficiency of stock prices decrease (Gresse 2017; Degryse, De Jong, and van Kervel 2015).

Second, this paper contributes to the broad literature on the flow of earnings information to markets. Prior research has examined characteristics that affect the flow of earnings information to markets, such as the information environment (Abarbanell and Bernard 1992; Roulstone 2003), trading by sophisticated investors (Utama and Cready 1997; Jiambalvo, Rajgopal, and Venkatachalam 2002; Ali, Klasa, and Li 2008), disclosure quality (Lee 2012), reporting regimes (Landsman, Maydew, and Thornock 2012), and information acquisition by investors (Drake, Roulstone, and Thornock 2015). We add to this literature by focusing on the role of dark pool trading.

Of particular note in this arena is recent work showing that algorithmic trading can affect the flow of information to markets. Weller (2018) and Lee and Watts (2021) argue that algorithmic traders are better than other traders at detecting informed investors, and can more rapidly incorporate informed investors' information into stock prices. This, in turn, increases the cost of trading on acquired information. The argument for why decreases in dark pool trading affect information acquisition also relates to the cost of trading, but the market structure underlying it is fundamentally different. While algorithmic trading is a standard, nearly universal investment technology, dark pool trading is an optional market-design tool. By focusing on dark pool trading, we are examining whether a specific market design can affect the level of information acquisition.

1. Data and Measures of Information Acquisition

We obtain weekly dark pool trading volume from FINRA's ATS Transparency Data. As part of FINRA's effort to increase market transparency, FINRA Rule 4552 requires each alternative trading system (ATS) to report its weekly volume information to FINRA on a stock-by-stock basis since the week of May 11, 2014. FINRA's ATS Transparency Data publishes the weekly volume information regarding Tier 1 NMS stocks, Tier 2 NMS stocks, and over-the-counter stocks. ATSs include electronic communications networks (ECNs) and dark pools. Since ECNs publicly display their quotations, they are generally considered lit markets. We exclude trades on ECNs, such as Bloomberg

⁸ See Buti, Rindi, and Werner (2011), Fleming and Nguyen (2013), Comerton-Forde and Putninš (2015), Foley and Putninš (2016), Hatheway, Kwan, and Zhen (2017), and Albuquerque, Song, and Yao (2020).

Tradebook, Citigroup Lava Flow, and Credit Suisse Light Pool.⁹ Overall, dark pools account for a substantial fraction of the total trading volume on U.S. equity markets: about 14% of the consolidated trading volume is executed on dark pools each month.

We obtain dark pool trading volume for common stocks listed on the NYSE, NYSE MKT, and NASDAQ (Center for Research in Security Prices, or CRSP, share codes 10 and 11; exchange codes 1, 2, and 3) from FINRA from July 2014 to December 2017. The sample period spans September 2014 through December 2017 to allow for a three month measurement window for quarterly measures of dark pool trading.

We combine the initial dark pool trading data with stock returns, trading volume, stock prices, and total shares outstanding from CRSP; consolidated trade data from Trade and Quote (TAQ); financial statement data from Compustat; analyst coverage and consensus forecasts from I/B/E/S; and institutional holding data from the Thomson-Reuters Institutional Holdings (13F) database.¹⁰ We exclude observations with missing values for the variables required in the regression analyses.

We use two measures to quantify information acquisition prior to quarterly earnings announcements: Weller's (2018) jump ratio and the future earnings response coefficient. The final sample consists of 40,150 firm-quarter observations for the analysis on the jump ratio, and 40,186 firm-quarter observations for the analysis on the future earnings response coefficient.

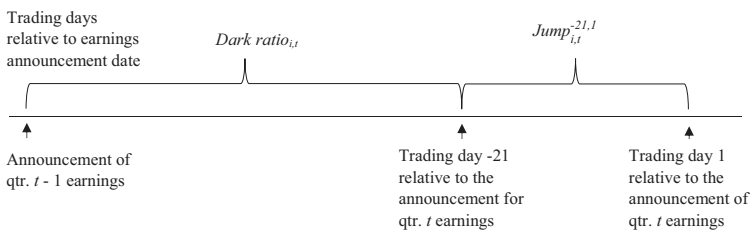
The jump ratio, $Jump_{i,t}^{-21,1}$, is defined as the ratio of cumulative abnormal return over trading days $[-1, 1]$ around earnings announcements ($CAR_{i,t}^{-1,1}$) divided by cumulative abnormal return over trading days $[-21, 1]$ relative to earnings announcements ($CAR_{i,t}^{-21,1}$). We compute daily abnormal return as the raw return less the buy-and-hold return to a benchmark portfolio of firms matched on size and book-to-market ratio over the same period. The benchmark portfolios are constructed using Fama and French's (1992) methodology. We classify all firms with CRSP share codes 10 and 11 into 25 portfolios by size at the end of June of year t and by the book-to-market ratio at the end of December of year $t - 1$.

The jump ratio measure captures the preemption of earnings news. Investors acquire information on upcoming earnings news before earnings announcements take place (Demski and Feltham 1994; McNichols and Trueman 1994; Ke and Petroni 2004; Vega 2006; Christophe, Ferri, and Hsieh 2010). When investors trade on their acquired information, this information becomes at least partially revealed to market participants, resulting in stock price adjustments in the direction of the acquired information prior to earnings

⁹ We are grateful to the Division of Trading and Markets of the SEC for providing the list of ECNs. More detailed information on ECNs is available at <https://www.sec.gov/divisions/marketreg/mr-noaction.shtml#ecns>.

¹⁰ The consolidated volume reported by the CRSP contains the trading volume executed on different market centers, including exchanges, ECNs, ATSS, and broker-dealers (Phillips 2009).

A Sample for event study



B Sample for FERC analysis

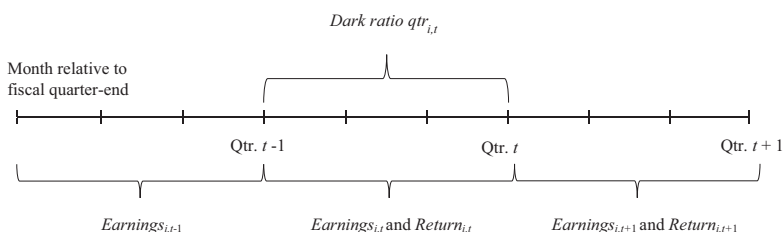


Figure 1
Sample construction timeline

announcements (Kyle 1985; Glosten and Milgrom 1985). Consequently, earnings announcements provide less new information and result in smaller price adjustments (Kim and Verrecchia 1991, 1997). A lower jump ratio implies that more of the total acquirable private information is acquired and incorporated into the stock price in the preannouncement period.

Figure 1, panel A, displays how we measure the time windows used in generating the jump ratio variable. We measure the level of dark pool trading during the preannouncement period of quarter t ($Dark\ ratio_{i,t}$). The preannouncement period includes all weeks between the earnings announcement date of quarter $t-1$ and 21 trading days before the announcement date of quarter t .

The second measure we use to quantify information acquisition is the future earnings response coefficient (FERC). Following Lundholm and Myers (2002), Ettredge et al. (2005), Glosten, Nallareddy, and Zou (2020), and Israeli, Lee, and Sridharan (2017), we estimate an augmented FERC model that includes dark pool trading and the interaction terms between dark pool trading and past earnings, dark pool trading and concurrent earnings, and dark pool trading and future earnings:

$$\begin{aligned}
 Return_{i,t} = & a_0 + a_1 \times Earnings_{i,t-1} + a_2 \times Earnings_{i,t} + a_3 \times Earnings_{i,t+1} \\
 & + a_4 \times Dark\ ratio\ qtr_{i,t} + a_5 \times Earnings_{i,t-1} \times Dark\ ratio\ qtr_{i,t}
 \end{aligned}$$

$$+a_6 \times \text{Earnings}_{i,t} \times \text{Dark ratio } qtr_{i,t} + a_7 \times \text{Earnings}_{i,t+1} \\ \times \text{Dark ratio } qtr_{i,t} + \text{Controls}_{i,t} + \varepsilon_{i,t}. \quad (1)$$

The variable $\text{Return}_{i,t}$ is the buy-and-hold return over month $t-2$ to month t relative to the fiscal quarter end. The variable $\text{Earnings}_{i,t+k}$ is quarterly seasonally adjusted net income before extraordinary items scaled by the market value of equity at the beginning of quarter $t+k$. The variable $\text{Dark ratio } qtr_{i,t}$ is the ratio of trading volume of firm i executed on dark pools divided by the consolidated volume in month $t-2$ to month t relative to the fiscal quarter end. Figure 1, panel B, depicts how we measure the time windows used in generating the FERC variable.

The coefficient for future earnings (a_3) is the traditional FERC. It captures the extent to which current stock returns reflect future earnings. In our setup therefore, a positive coefficient for the interaction term $\text{Earnings}_{i,t+1} \times \text{Dark ratio } qtr_{i,t}$ (a_7) indicates that dark pool trading is associated with a higher FERC. That is, an increase in dark pool trading is accompanied by an increase in the amount of future earnings information that is acquired and reflected in current returns.

Prior research investigates the role of sophisticated market participants on the FERC. Certain economic actions taken by a firm in the current period (e.g., long-term sales contracts and investment activities) affect future earnings. By acquiring and processing information about these actions, sophisticated investors, such as institutional investors (Ayers and Freeman 2003), insiders (Crawford, Park, and Roulstone 2011), and short sellers (Drake et al. 2015), become informed about future earnings. If information about future earnings is acquired and incorporated into prices, current stock prices will reflect more information about future earnings.

Table 1 reports summary statistics for the sample. The average stock has 14% of its total trading volume executed in dark pools. The mean (median) jump ratio over trading days $[-21, 1]$, $\text{Jump}_{i,t}^{-21,1}$, is 0.420 (0.412). The average (median) firm has a market capitalization of \$6.965 billion (\$0.785 billion), suggesting that the distribution of market capitalization in the sample is skewed to the right. The average firm in the sample has 7.186 analysts following it and 61.8% institutional ownership. Table 1 reports summary statistics for $\text{Return}_{i,t}$ and $\text{Earnings}_{i,t}$. However, since the FERC itself is the result of a regression, we cannot provide summary statistics.

2. Relation between Dark Pool Trading and Information Acquisition

We begin by examining whether dark pool trading increases the preemption of earnings news. We estimate the following regression:

$$\text{Jump}_{i,t}^{-21,1} = \alpha + \beta \text{Dark Ratio}_{i,t} + \gamma' X_{i,t} + \theta_i + \theta_t + \varepsilon_{i,t}, \quad (2)$$

Table 1
Summary statistics

Variable	N	Mean	SD	10%	25%	50%	75%	90%
Dark pool trading variables								
<i>Dark ratio</i> _{<i>i,t</i>}	40,150	0.141	0.059	0.053	0.105	0.147	0.181	0.212
<i>Dark ratio qtr</i> _{<i>i,t</i>}	40,186	0.143	0.059	0.054	0.107	0.148	0.182	0.212
Other variables								
<i>Jump</i> _{<i>i,t</i>} ^{-21,1}	40,150	0.420	2.873	-0.956	-0.061	0.412	0.912	1.790
<i>Return</i> _{<i>i,t</i>}	40,186	0.017	0.209	-0.225	-0.088	0.017	0.116	0.244
<i>Earnings</i> _{<i>i,t</i>}	40,186	-0.002	0.067	-0.028	-0.006	0.001	0.006	0.023
<i>Size</i> _{<i>i,t</i>} (\$ billions)	40,150	6.965	28.954	0.048	0.166	0.785	3.297	12.290
<i>Book-to-market</i> _{<i>i,t</i>}	40,150	0.584	0.510	0.123	0.248	0.469	0.779	1.124
<i>Leverage</i> _{<i>i,t</i>}	40,150	0.182	0.189	0.000	0.008	0.129	0.299	0.450
<i>Growth</i> _{<i>i,t</i>}	40,150	0.018	0.132	-0.077	-0.024	0.005	0.033	0.087
<i>Idiosyncratic volatility</i> _{<i>i,t</i>}	40,150	0.019	0.013	0.007	0.010	0.015	0.024	0.038
<i>#Analysts</i> _{<i>i,t</i>}	40,150	7.186	7.418	0.000	2.000	5.000	10.000	18.000
<i>Institutional ownership</i> _{<i>i,t</i>}	40,150	0.618	0.335	0.067	0.340	0.711	0.894	0.985
<i>Loss</i> _{<i>i,t</i>}	40,150	0.336	0.472	0.000	0.000	0.000	1.000	1.000
<i>Spread</i> _{<i>i,t</i>}	40,150	0.010	0.007	0.004	0.005	0.008	0.012	0.019
<i>Spread</i> _{<i>i,t</i>} ^{-21,1}	1,465	0.009	0.005	0.004	0.006	0.008	0.011	0.016
<i>Odd-lot ratio</i> _{<i>i,t</i>} ^{-21,1}	1,465	0.190	0.102	0.081	0.122	0.172	0.236	0.309
<i>Trade-to-order ratio</i> _{<i>i,t</i>} ^{-21,1}	1,465	0.038	0.020	0.013	0.022	0.036	0.050	0.063
<i>Cancel-to-trade ratio</i> _{<i>i,t</i>} ^{-21,1}	1,465	34.591	33.799	11.425	15.641	23.725	40.821	65.046
<i>Average trade size</i> _{<i>i,t</i>} ^{-21,1}	1,465	0.101	0.040	0.066	0.076	0.089	0.112	0.152
<i>Price delay</i> _{<i>i,t</i>} ^{-21,1}	1,465	1.358	0.531	0.727	0.962	1.301	1.692	2.081
<i>Abs(Variance ratio)</i> _{<i>i,t</i>} ^{-21,1}	1,465	0.581	0.795	0.071	0.184	0.373	0.608	1.246
<i>Autocorrelation</i> _{<i>i,t</i>} ^{-21,1}	1,465	-0.101	0.239	-0.416	-0.274	-0.103	0.066	0.211
<i>#10 - K</i> _{<i>i,t</i>} ^{-21,-2}	1,448	0.070	0.201	0.003	0.005	0.012	0.038	0.147
<i>#10 - Q</i> _{<i>i,t</i>} ^{-21,-2}	1,448	0.068	0.256	0.002	0.003	0.009	0.030	0.132
<i>#Other filings</i> _{<i>i,t</i>} ^{-21,-2}	1,448	6.067	17.624	0.237	0.468	1.212	3.843	13.524
<i>Return</i> _{<i>i,t</i>} ^{-1,18}	1,136	0.030	0.117	-0.131	-0.040	0.026	0.099	0.192
<i>CAR</i> _{<i>i,t</i>} ^{-1,18}	1,136	0.005	0.114	-0.147	-0.067	0.002	0.074	0.166
<i>Hedge fund holdings</i> _{<i>i,t</i>}	1,459	0.089	0.082	0.009	0.032	0.067	0.118	0.196
<i>Price impact off-exchg.</i> _{<i>i,t</i>} ^{-21,-2}	1,446	0.033	1.665	-0.113	-0.003	0.006	0.030	0.164
<i>Synchronicity</i> _{<i>i,t</i>} ^{-21,-2}	1,415	-0.971	1.425	-2.798	-1.838	-0.852	-0.014	0.742
<i>Short ratio</i> _{<i>i,t</i>} ^{-21,-2}	1,136	0.382	0.132	0.211	0.287	0.381	0.475	0.549

This table presents univariate statistics for key variables in the sample. Table A.1 in the appendix defines the variables.

where the subscripts *i* and *t* denote firm *i* and year-quarter *t*, respectively. The variable *Jump*_{*i,t*}^{-21,1} is the jump ratio defined in Section 1.

The variable *Dark ratio*_{*i,t*} is the ratio of the trading volume of firm *i* executed on dark pools divided by the consolidated volume during the preannouncement period of quarter *t*. The preannouncement period includes all weeks between the earnings announcement date of quarter *t* - 1 and 21 trading days before the announcement date of quarter *t*. If an earnings announcement for quarter *t* - 1 is released on a Friday, we exclude the week following the earnings announcement. If dark pool trading increases information acquisition in the preannouncement period, we expect to see a negative coefficient for *Dark ratio*_{*i,t*} (*β*) in Equation (2).

The vector $X_{i,t}$ is a set of control variables that prior research has found to be associated with price reactions to earnings news. To control for cross-sectional differences in the riskiness of firms, we include firm size ($\log(\text{Size})_{i,t}$), book-to-market ratio ($\text{Book-to-market}_{i,t}$), and leverage ($\text{Leverage}_{i,t}$) (Fama and French 1992). To control for firm-specific risk, we include idiosyncratic return volatility ($\text{Idiosyncratic volatility}_{i,t}$). We include analyst coverage ($\log(\#\text{Analysts})_{i,t}$) and institutional ownership ($\text{Institutional ownership}_{i,t}$), as these intermediaries can affect the information environment (Utama and Cready 1997; Ali, Klasa, and Li 2008). The information environment can affect both the dissemination of earning news and the level of dark trading (Boulton, Braga-Alves, and Chakrabarty 2020; Balakrishnan et al. 2020). We include the Corwin and Schultz (2012) effective spread measure ($\text{Spread}_{i,t}$) to control for liquidity. Finally, we include firm fixed effects, θ_i , and year-quarter fixed effects, θ_t , to control for unobserved heterogeneity over time and across firms. We cluster standard errors by firm and earnings announcement date (Hirshleifer, Lim, and Teoh 2009).

Table 2 provides the results from estimating Equation (2). Column 1 reports the results without year-quarter fixed effects, and column 2 shows the results after controlling for both firm and year-quarter fixed effects. In columns 1 and 2, we find that the coefficient for $\text{Dark ratio}_{i,t}$ is statistically significant and negative. This suggests that when the level of dark pool trading is high, more information is acquired before earnings announcements. In terms of economic significance, the jump ratio decreases by 23% $([-0.776 \times 0.147] - (-0.776 \times 0.181)) / (-0.776 \times 0.147)$ if a firm's *Dark ratio* increases from a 50th percentile value of 0.147 to a 75th percentile value of 0.181.¹¹

As for the control variables, we find little statistical significance. Our prior is that firms with less information asymmetry, such as those with more analysts or firms that are larger, would have a lower jump ratio. However, the lack of statistical significance is not surprising as we include firm fixed effects in the regression specifications. The firm fixed effects soak up the time invariant levels of the independent variables. In addition, the 3.5-year sample is not a great deal of time for firms to experience times-series changes in these variables.

Next, we test whether dark pool trading affects longer horizon information acquisition. If dark pool trading increases information acquisition, then the association between current stock returns and future earnings should increase. That is, dark pool trading should lead to an increase in the mapping of future earnings to current returns—the FERC. We estimate the following regression

¹¹ The adjusted *R*-squared values in both specifications are low, at .004. Recall that the jump ratio is the ratio of two abnormal return measures. As such we are essentially explaining abnormal returns. This is a notoriously difficult task, and low-adjusted *R*-squared values are not uncommon. Even though the model does not capture much of the dependent variable's variability, the individual covariates' coefficients and *t*-statistics still measure the relation between the independent variable and the dependent variable.

Table 2
Effect of dark pool trading on the process of price formation prior to earnings announcements

Dependent variable =	$Jump_{i,t}^{-21,1}$	
	(1)	(2)
<i>Dark ratio</i> _{<i>i,t</i>}	−0.852** (−2.00)	−0.776* (−1.79)
<i>log(Size)</i> _{<i>i,t</i>}	0.037 (0.67)	0.031 (0.55)
<i>Book-to-market</i> _{<i>i,t</i>}	0.024 (0.33)	0.019 (0.25)
<i>Leverage</i> _{<i>i,t</i>}	−0.361 (−1.60)	−0.386* (−1.70)
<i>Idiosyncratic volatility</i> _{<i>i,t</i>}	−3.626* (−1.73)	−4.733** (−2.22)
<i>log(#Analysts)</i> _{<i>i,t</i>}	0.021 (0.29)	0.021 (0.28)
<i>Institutional ownership</i> _{<i>i,t</i>}	−0.042 (−0.25)	−0.086 (−0.51)
<i>Spread</i> _{<i>i,t</i>}	−0.159 (−0.04)	−1.139 (−0.28)
Year-quarter FEs	No	Yes
Firm FEs	Yes	Yes
Observations	40,150	40,150
Adjusted <i>R</i> -squared	.004	.004

This table reports the estimation results of regressing the jump ratio ($Jump_{i,t}^{-21,1}$) on dark pool trading (*Dark ratio*_{*i,t*}) from Equation (3). $Jump_{i,t}^{-21,1}$ is the ratio of cumulative abnormal returns over trading days [−1, 1] around earnings announcements, $CAR_{i,t}^{-1,1}$, divided by the cumulative abnormal returns over trading days [−21, 1] around earnings announcements, $CAR_{i,t}^{-21,1}$. The variable *Dark ratio*_{*i,t*} is the ratio of the trading volume of firm *i* executed on dark pools, divided by the consolidated volume during the preannouncement period of quarter *t*. For brevity, the intercept, firm fixed effects, and year-quarter fixed effects are not reported. Standard errors are clustered by firm and earnings announcement date. Table A.1 in the appendix defines the variables. **p* < .1; ***p* < .05; ****p* < .01.

specification using quarterly data (Glosten, Nallareddy, and Zou 2020; Israeli, Lee, and Sridharan 2017; Lundholm and Myers 2002; Ettredge et al. 2005):

$$Return_{i,t} = \alpha + \sum_{k=-1}^1 [\delta_k Earnings_{i,t+k} \times Dark\ ratio\ qtr_{i,t} + \theta_k Earnings_{i,t+k} + \psi' Earnings_{i,t+k} \times X_{i,t}] + \beta_2 Dark\ atio\ qtr_{i,t} + \gamma' X_{i,t} + \theta_i + \theta_t + \varepsilon_{i,t}, \tag{3}$$

where the subscripts *i* and *t* denote firm *i* and year-quarter *t*, respectively. The subscript *k* takes the values −1, 0, and 1 to allow for the relevant variables in different time periods to be represented in the equation. As in Equation (1), the variable *Return*_{*i,t*} is the buy-and-hold return over month *t* − 2 to month *t* relative to the fiscal quarter end. The variable *Earnings*_{*i,t+k*} is quarterly seasonally adjusted net income before extraordinary items scaled by the market value of equity at the beginning of quarter *t* + *k*. *k* iterates between −1, 0, and 1 so that the three different periods of earnings with subscript *k* are captured in the specification. The variable *Dark ratio qtr*_{*i,t*} is the ratio of trading volume of

firm i executed on dark pools divided by the consolidated volume in month $t - 2$ to month t relative to the fiscal quarter end.

The variable of interest is the interaction term, $Earnings_{i,t+1} \times \text{Dark ratio } qtr_{i,t}$. A positive coefficient δ_1 on the interaction term $Earnings_{i,t+1} \times \text{Dark ratio } qtr_{i,t}$ indicates that the FERC is higher for firms with a higher level of $\text{Dark ratio } qtr_{i,t}$.

To control for factors that prior research finds to be associated with the FERC, we use the vector $X_{i,t}$ to denote a set of control variables, which includes $Return_{i,t+1}$, $\log(\text{Size})_{i,t}$, $\text{Book-to-market}_{i,t}$, $\text{Leverage}_{i,t}$, $\text{Growth}_{i,t}$, $\text{Idiosyncratic volatility}_{i,t}$, $\log(\#\text{Analysts})_{i,t}$, $\text{Spread}_{i,t}$, $\text{Institutional ownership}_{i,t}$, and $\text{Loss}_{i,t}$ (Lundholm and Myers 2002; Choi et al. 2011). Specifically, the buy-and-hold return in quarter $t + 1$ ($Return_{i,t+1}$) controls for the potential measurement error induced by using actual future earnings as a proxy for expected value (Collins et al. 1994). Because high-growth firms tend to have larger FERCs, we include book-to-market ratio ($\text{Book-to-market}_{i,t}$) and growth in assets ($\text{Growth}_{i,t}$). We include idiosyncratic return volatility ($\text{Idiosyncratic volatility}_{i,t}$) to control for firm-specific uncertainty, and firm size ($\log(\text{Size})_{i,t}$), analyst coverage ($\log(\#\text{Analysts})_{i,t}$), and institutional ownership ($\text{Institutional ownership}_{i,t}$) to control for cross-sectional differences in firms' information environments. Because negative future earnings may be more difficult to predict than positive future earnings, we follow Lundholm and Myers (2002) and Ettredge et al. (2005) and include an indicator variable, $\text{Loss}_{i,t}$, which takes the value one if firm i had negative earnings in quarter $t + 1$ and zero otherwise. Firms with negative earnings are expected to have lower FERCs. We also include the Corwin and Schultz (2012) effective spread measure, $\text{Spread}_{i,t}$, to control for the impact of liquidity. These variables are measured at the end of the fiscal quarter. We account for time-invariant firm heterogeneity and time effects by including firm and year-quarter fixed effects. We cluster standard errors by firm and year-quarter.

Table 3 presents the results from estimating Equation (3). Columns 1 and 2 report results for the tests without interactions between earnings and control variables. Columns 3 and 4 show results after controlling for those interactions. In all columns we find that the coefficient for the interaction term $Earnings_{i,t+1} \times \text{Dark ratio } qtr_{i,t}$ is statistically significant and positive. The results suggest that an increase in the level of dark pool trading leads to an increase in the association between contemporaneous returns and future earnings.

Overall, the results show that a higher level of dark pool trading leads to an increase in longer horizon information acquisition.

3. Causal Relation between Dark Pool Trading and Information Acquisition

To identify dark pools' effect on information acquisition, we use the SEC's Tick Size Pilot Program to perform a DiD analysis. The 2012 Jumpstart Our Business Startups (JOBS) Act directed the SEC to assess how decimalization affects the

Table 3
Effect of dark pool trading on FERCs

Dependent variable =	<i>Return_{i,t}</i>			
	(1)	(2)	(3)	(4)
<i>Earnings_{i,t+1}</i> × <i>Dark ratio qtr_{i,t}</i>	1.377*** (4.63)	1.237*** (4.26)	0.822** (2.21)	0.716* (1.92)
<i>Earnings_{i,t}</i> × <i>Dark ratio qtr_{i,t}</i>	−0.001 (−0.00)	−0.174 (−0.43)	0.040 (0.07)	−0.056 (−0.10)
<i>Earnings_{i,t−1}</i> × <i>Dark ratio qtr_{i,t}</i>	−0.233 (−0.55)	−0.217 (−0.55)	0.605 (1.08)	0.692 (1.32)
<i>Earnings_{i,t+1}</i>	−0.019 (−0.44)	−0.032 (−0.78)	−0.239* (−1.90)	−0.226* (−1.85)
<i>Earnings_{i,t}</i>	0.216*** (3.81)	0.187*** (3.38)	0.333*** (2.12)	0.418*** (2.71)
<i>Earnings_{i,t−1}</i>	−0.008 (−0.16)	−0.006 (−0.12)	0.135 (0.83)	0.224 (1.47)
<i>Return_{i,t+1}</i>	−0.118*** (−16.70)	−0.130*** (−18.24)	−0.114*** (−16.07)	−0.127*** (−17.75)
<i>Dark ratio qtr_{i,t}</i>	−0.305*** (−9.68)	−0.229*** (−7.49)	−0.295*** (−9.41)	−0.219*** (−7.21)
<i>Controls</i>	Yes	Yes	Yes	Yes
<i>Earnings_{i,t−1}</i> × <i>Controls</i>	No	No	Yes	Yes
<i>Earnings_{i,t}</i> × <i>Controls</i>	No	No	Yes	Yes
<i>Earnings_{i,t+1}</i> × <i>Controls</i>	No	No	Yes	Yes
Year-quarter FEs	No	Yes	No	Yes
Firm FEs	Yes	Yes	Yes	Yes
Observations	40,186	40,186	40,186	40,186
Adjusted <i>R</i> -squared	.066	.143	.074	.150

This table reports the estimation results of regressing current quarterly stock returns (*Return_{i,t}*) on future earnings (*Earnings_{i,t+1}*), the interaction between future earnings and dark pool trading (*Earnings_{i,t+1}* × *Dark ratio qtr_{i,t}*), control variables (*Controls_{i,t}*), and the interaction terms. The variable *Return_{i,t}* denotes quarterly buy-and-hold return of firm *i* for quarter *t*. The variable *Earnings_{i,t}* is quarterly seasonally adjusted net income before extraordinary items, scaled by the market value of equity at the beginning of quarter *t*. The variable *Dark ratio qtr_{i,t}* denotes the ratio of the trading volume executed in dark pools divided by the consolidated volume in month *t* − 2 to month *t* relative to the fiscal quarter end. In column 2, *Dark ratio_{i,t}* is continuous. For brevity, the intercept, the interaction terms *Earnings_{i,t−1}* × *Controls_{i,t}*, *Earnings_{i,t}* × *Controls_{i,t}*, and *Earnings_{i,t+1}* × *Controls_{i,t}*, firm fixed effects, and year-quarter fixed effects are not reported. Standard errors are clustered by firm and year-quarter. Table A1 in the appendix defines the variables. **p* < .1; ***p* < .05; ****p* < .01.

liquidity and trading of smaller-capitalization companies. In response to this request, the SEC launched the Tick Size Pilot Program on October 3, 2016, and implemented it on a staggered basis over the next two years. The pilot program consists of a control group of 1,200 randomly selected stocks and three treatment groups of 400 stocks each.¹²

The three treatment groups are subject to different changes to the trading environment. The stocks in the control group of the Tick Size Pilot Program continue quoting and trading in 1-cent increments. The stocks in treatment group 1 (G1) must be quoted in 5-cent increments. In addition to being subject to the rules applying to G1 stocks, the stocks in treatment group 2 (G2) must be traded in 5-cent increments. In addition to being subject to the rules applying to G2 stocks, the stocks in treatment group 3 (G3) are subject to the “trade-at” provision, which requires that orders be executed in lit venues unless dark

¹² To facilitate random assignment, the SEC selected stocks by a stratified sampling based on market capitalization, volume-weighted average price, and average daily volume.

venues can execute them at a meaningfully better price. By increasing the costs of trading in dark pools, the “trade-at” provision should decrease the trading volume in those pools.

In the DiD analysis, following Farley, Kelley, and Puckett (2018), we use G3 stocks as the treatment stocks and G2 stocks as the control stocks. A comparison of effects between G2 and G3 holds constant the quoting and trading increments and thereby isolates any dark pool trading effects.¹³ We use a $[-120, 120]$ -day window around the effective dates of the pilot program to conduct the DiD analysis. This narrow window helps control for any unobserved variables, since unobserved variables are less likely to change significantly during such a short period.

Before performing the DiD analysis, we document that the treatment stocks that are subject to the “trade-at” provision experience a decrease in dark pool trading volume. Figure 2 plots the ratio of the consolidated trading volume executed on dark pools (*Dark ratio*) by week for G2 stocks (control stocks) and G3 stocks (treatment stocks) over weeks $t-20$ to $t+20$ relative to the implementation of the Tick Size Pilot Program.

The figure shows that control stocks and treatment stocks have a similar level of dark pool trading prior to the pilot program’s implementation. Consistent with the idea that the “trade-at” provision should result in an increase in the trading costs in dark pools, treatment stocks experience a decrease in the level of dark pool trading from the implementation week of the pilot program.¹⁴ The average market share of dark pools for stocks in the G3 treatment group drops by 30% (from 14.6% to 10.3%) from the 20 weeks before to the 20 weeks after the pilot program implementation dates.

The validity of the DiD estimator depends on the parallel trend assumption, which is that the underlying trend in the outcome variable is similar between

¹³ A number of recent studies use the Tick Size Pilot Program to examine other effects. Farley, Kelley, and Puckett (2018) examine the implication of dark pool trading on trading costs and informational efficiency of stock price. As in our study, they use G3 stocks as the treatment stocks and G2 stocks as the control stocks. They find that following the decrease in dark trading after the implementation of the Tick Size Pilot Program, trading costs remain unchanged, and the informational efficiency of stock prices change minimally. A number of studies use the pilot program to study the effects of tick size on various issues (see Rindi and Werner 2017; Albuquerque, Song, and Yao 2020; Li, Ye, and Zheng 2018; Lee and Watts 2021). Unlike our paper, these studies compare the effects between the control group and the treatment groups of the Pilot (G1, G2, and G3 stocks). Rindi and Werner (2017) show that the larger tick size causes spreads and market depth to increase. Albuquerque, Song, and Yao (2020) test the effect of the Tick Size Pilot Program on stock prices and show a decrease in stock prices following the Pilot. Li, Ye, and Zheng (2018) test the effect of tick size on corporate payout policies. Lee and Watts (2021) examine how an increase in tick size affects algorithmic trading and the price discovery process. Chakrabarty, Cox, and Upson (2021) find that the overall price discovery process, as measured by Hasbrouck’s (1995) information share, changed following the implementation of the Tick Size Pilot Program.

¹⁴ Figure 2 shows that the control group (G2) experiences an increase in dark pool trading. This is not surprising, given the forced minimum trading increment of 5 cents that applies to the G2 stocks. While dark pools are also subject to the 5-cent trade increment, they can avail themselves of the midpoint exemption and trade at the NBBO midpoint. Several dark pools execute orders at the midpoint of the NBBO. With the increase in the minimum trading increments on-exchange, these midpoint dark pools have become an attractive way to reduce transaction costs. The fact that the treatment and control groups move similarly in the preperiod suggests the parallel trends assumption is satisfied. As discussed, as the G3 stocks are subject to all the same rules as G2, using G2 as a control group is valid in a DiD analysis.

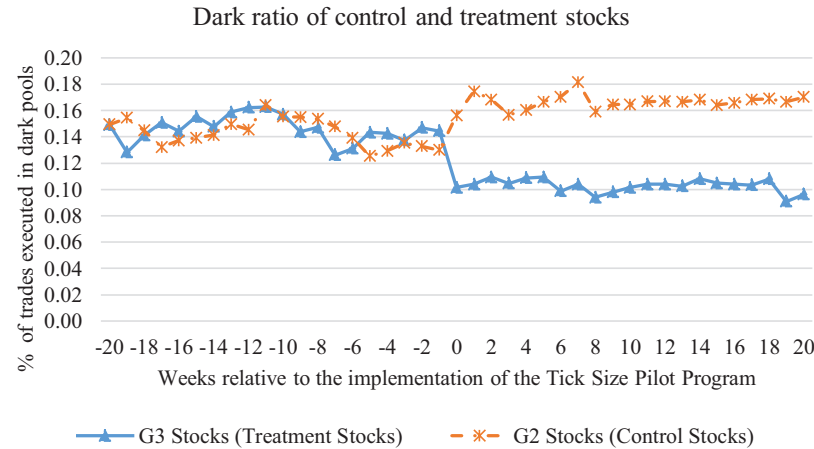


Figure 2
Differences in the ratio of the trading volume executed on dark pools for treatment and control stocks
This figure plots the ratio of the trading volume of common shares executed on dark pools (*Dark ratio*) by week over weeks $t-20$ to $t+20$ relative to the implementation of the Tick Size Pilot Program.

the treatment and control groups. To provide suggestive evidence of parallel trends, we perform a t -test on the differences in characteristics between the two groups prior to the implementation of the pilot program. Panel A of Table 4 shows that, of the 14 firm characteristics we examine, *log(Size)*, *Idiosyncratic volatility*, *Institutional ownership*, and *Spread* display a statistically significant difference between the treatment group and the control group.

Nonetheless, the economic differences between the two groups are small. For example, the average firm size is \$0.734 billion for the treatment group and \$0.635 billion for the control group. Although this difference is statistically different from zero, the economic difference between such values is unlikely to have much impact. Importantly, the two groups have a similar level of dark pool trading—14.4% for the treatment group and 14.8% for the control group—prior to the pilot program’s implementation.

For the analysis focusing on the jump ratio, we want to allow the pilot program to affect the level of dark pool trading and thus the price reaction during the preannouncement period and earnings announcements. To accomplish this, we exclude the earnings announcements made in the $[-1, 20]$ trading day window around the effective dates of the pilot program. We conduct the DiD analysis on the jump ratio by estimating the following specification:

$$Jump_{i,t}^{-21,1} = \alpha + \beta_1 Post_t \times Treatment_i + \beta_2 Post_t + \gamma' X_{i,t} + \theta_i + \theta_t + \varepsilon_{i,t}, \quad (4)$$

where the subscripts i and t refer to firm i and year-quarter t , respectively. The variable $Jump_{i,t}^{-21,1}$ is defined in Section 1. The indicator variable $Treatment_{i,t}$ equals one for treatment firms and zero for control firms. The indicator variable

Table 4
Difference-in-differences analysis

A. Differences in variables prior to the implementation of the Tick Size Pilot Program

Variable	Treatment	Control	Difference	t-statistic
Dark pool trading variables				
Dark ratio _{i,t}	0.144	0.148	−0.003	−0.85
Dark ratio qtr _{i,t}	0.147	0.148	−0.001	−0.40
Other variables				
Jump ^{−21,1} _{i,t}	0.376	0.619	−0.243	−1.19
Return _{i,t}	0.042	0.053	−0.010	−0.89
Earnings _{i,t}	0.000	0.000	0.001	0.19
Size _{i,t} (\$ billions)	0.734	0.635	0.098	1.89*
Book-to-market _{i,t}	0.656	0.704	−0.048	−1.35
Leverage _{i,t}	0.136	0.155	−0.019	−1.48
Growth _{i,t}	0.024	0.021	0.003	0.40
Idiosyncratic volatility _{i,t}	0.016	0.020	−0.003	−4.80***
#Analysts _{i,t}	3.899	4.091	−0.192	−0.78
Institutional ownership _{i,t}	0.581	0.619	−0.038	−1.69*
Loss _{i,t}	0.248	0.288	−0.040	−1.28
Spread _{i,t}	0.009	0.011	−0.002	−4.97***

B. Estimation results of the difference-in-differences analysis

Dependent variable =	Jump ^{−21,1} _{i,t}	Return _{i,t}
	(1)	(2)
Post _{i,t} × Treatment _{i,t}	0.788** (2.35)	0.002 (0.08)
Earnings _{i,t+1} × Post _{i,t} × Treatment _{i,t}		−2.503** (−2.40)
Post _{i,t}	−1.060** (−2.14)	−0.004 (−0.08)
Earnings _{i,t+1} × Treatment _{i,t}		0.938** (1.98)
Earnings _{i,t+1} × Post _{i,t}		2.153** (2.45)
Controls	Yes	Yes
Earnings _{i,t−1} × Controls	N/A	Yes
Earnings _{i,t} × Controls	N/A	Yes
Earnings _{i,t+1} × Controls	N/A	Yes
Year-quarter FEs	Yes	Yes
Firm FEs	Yes	Yes
Observations	1,465	1,459
Adjusted R-squared	.008	.096

Panel A reports variable averages for the treatment and control groups prior to the implementation of the pilot program, the differences in means of each variable, and the corresponding *t*-statistics. Panel B provides the estimation results of the DiD regression specifications in Equations (4) and (5). Table A.1 in the appendix defines the variables. Standard errors are clustered by firm and earnings announcement date in column 1 and are clustered by firm and year-quarter in column 2. **p* < .1; ***p* < .05; ****p* < .01.

Post_{i,t} equals one for dates after the pilot program is implemented for stock *i* and zero otherwise.

In Equation (4), the coefficient for the interaction term Post_{i,t} × Treatment_{i,t} (β₁) captures the changes in information acquisition taking place in the preannouncement period for treatment firms relative to control firms. We expect the decrease in the level of dark pool trading resulting from the implementation of the pilot program to result in a smaller preemption of upcoming earnings

news. That is, we expect the sign on β_1 in Equation (4) to be positive. The vector $X_{i,t}$ is the same set of control variables as in Equation (2). We account for time-invariant firm heterogeneity and time effects by including firm fixed effects, θ_i , and year-quarter fixed effects, θ_t . We cluster standard errors by firm and earnings announcement date. Table 4, panel B, reports the results.

Column 1 presents the results of the DiD analysis based on Equation (4). The coefficient for the interaction term $Post_{i,t} \times Treatment_{i,t}$ is positive and statistically significant, indicating a smaller preemption of upcoming earnings news for treated firms than for control firms after the Tick Size Pilot Program is introduced. The result is consistent with a decrease in the level of dark pool trading reducing the amount of information that is acquired before earnings announcements. It is worth noting that the coefficient for $Post_{i,t}$ is negative and statistically significant, consistent with the findings of Lee and Watts (2021) that G1, G2, and G3 overall experienced a decline in algorithmic trading, resulting in great information acquisition.

We repeat the DiD analysis on the FERC measure of information acquisition. For each firm in the sample, we exclude the fiscal quarter when the pilot program is implemented. The regression specification is:

$$\begin{aligned}
 Return_{i,t} = & \alpha + \sum_{k=-1}^1 [\delta_k Earnings_{i,t+k} \times Post_{i,t} \times Treatment_{i,t} + \mu_k Earnings_{i,t+k} \\
 & \times Post_{i,t} + \phi_k Earnings_{i,t+k} \times Treatment_{i,t} + \beta_k Earnings_{i,t+k} \\
 & + \psi' Earnings_{i,t+k} \times X_{i,t}] + \beta_2 Post_t \times Treatment_i + \gamma' X_{i,t} \\
 & + \theta_i + \theta_t + \varepsilon_{i,t},
 \end{aligned} \tag{5}$$

where the subscripts i and t refer to firm i and year-quarter t , respectively. $Return_{i,t}$ is defined in Section 2. The indicator variable $Treatment_{i,t}$ equals one for treatment firms, and zero for control firms. The indicator variable $Post_{i,t}$ equals one for dates after the pilot program is implemented. The vector $X_{i,t}$ is the same set of control variables as in Equation (3). The coefficient for the interaction term $Earnings_{i,t+1} \times Post_{i,t} \times Treatment_{i,t} (\delta_1)$ captures the change in the FERC for treatment firms relative to control firms. We expect the decrease in the level of dark pool trading resulting from the implementation of the pilot program to be associated with a smaller FERC. That is, we expect a negative coefficient for δ_1 in Equation (5). Similar to Equation (4), we include firm fixed effects, θ_i , and year-quarter fixed effects, θ_t . We cluster standard errors by firm and year-quarter.

Column 2 of Table 4, panel B, reports the results. We find a smaller FERC for treated firms than for control firms after the introduction of the Tick Size Pilot Program, as evidenced by a negative and statistically significant coefficient for

the interaction term $Earnings_{i,t+1} \times Post_{i,t} \times Treatment_{i,t}$. Overall, these results suggest that dark pool trading increases information acquisition.^{15, 16}

4. Further Evidence of Changes in Information Acquisition

To provide further evidence that dark pool trading increases information acquisition, we test whether we can detect changes in information acquisition. The first test aims to detect directly the amount of information acquired, as shown by investors' accessing SEC EDGAR reports. We directly identify a statistically significant decrease in information acquisition, as SEC EDGAR searches decrease along with decreases in dark pool trading. The second test aims to pin down whether the acquired information is marketwide or firm specific. We find that dark pool trading encourages the acquisition of firm-specific information. Taken together, these results provide more nuanced evidence of dark pool trading affecting information acquisition.

4.1 Direct evidence of information acquisition via EDGAR

The jump ratio and FERC measures of information acquisition are well established in the literature and supported by a clear economic rationale. Nonetheless, a measure that tracks the actual activities of information acquisition might be even more convincing. For such a measure, we turn to the SEC's EDGAR database. Recent work shows that financial information disclosed by the EDGAR database is acquired by sophisticated market participants, such as financial analysts (Gibbons, Iliev, and Kalodimos 2020) and institutional investors (Chen et al. 2020; Crane, Crotty, and Umar 2018), as well as by investors in general (Lee, Ma, and Wang 2015; Loughran and McDonald 2017; Drake, Roulstone, and Thornock 2015). Following these studies, we measure information acquisition directly by tracking SEC EDGAR database usage.

¹⁵ Recall from Figure 2 that G2, the control group, experienced an increase in dark pool trading (because lit exchanges were required to trade at \$0.05 increments but dark venues were not). It is noteworthy that in the jump ratio analysis the coefficient for *Post* is negative and statistically significant, consistent with an increase in dark pool trading increasing information acquisition. In Column (2) for the FERC analysis the coefficient for $Post \times Earnings$ is positive and statistically significant, also consistent with an increase in dark pool trading increasing information acquisition. However, we do not want to overweight these results, as they also can be driven by more general time series variation for reasons unrelated to dark pool trading. And while we observe these coefficients being statistically significant in the expected direction (if dark pool trading increased information acquisition), the same is not true for all the remaining tables. Although none of the coefficients in those tables is statistically significant in the wrong direction, the results are often not statistically different from zero.

¹⁶ In untabulated results, we follow Rindi and Werner (2019) and repeat our main analysis in Table 4, panel B, on two subsets of the data: a tick-constrained group and a tick-unconstrained group. The tick-constrained sample contains stocks with a preplot quoted spread of less than 5 cents. The coefficient of interest ($Post \times Treatment$ for the jump ratio and $Post \times Treatment \times Earnings$ for FERC) is statistically significant in the expected direction for the tick-constrained subsample and not statistically different from zero for the tick-unconstrained subsample, but a chi-square test between the two subsamples does not show a statistically significant difference between the two coefficients. This may partially be a power issue, as we are taking an already small number of observations and further subdividing them.

The EDGAR database hosts all of the regulatory filings mandated by the SEC, including annual and quarterly reports (10-K and 10-Q), current reports (8-K), equity ownership updates (Form 4), registration statements (S-1), and proxy statements (DEF 14A), among others. The SEC EDGAR server log files record and store user access statistics for regulatory filings.

Using the EDGAR server logs, we construct three measures of EDGAR usage: the number of downloads of 10-K filings (*#10-K*), 10-Q filings (*#10-Q*), and all other EDGAR filings (*#Other filings*). To capture preannouncement information acquisition, we compute, for each filing type, the average number of requests made in the period of trading days $[-21, -2]$ relative to earnings announcements, scaled by trading volume, and finally multiplied by 100. We estimate Equation (6) with the dependent variable *#Report* representing one of three measures based on EDGAR search volume, $\#10-K_{i,t}^{-21,-2}$, $\#10-Q_{i,t}^{-21,-2}$, or $\#Other\ filings_{i,t}^{-21,-2}$.

$$\#Report_{i,t} = \alpha + \beta_1 Post_t \times Treatment_i + \beta_2 Post_t + \gamma' X_{i,t} + \theta_i + \theta_t + \varepsilon_{i,t}. \quad (6)$$

If dark pool trading encourages information acquisition, then the treatment stocks (G3) should experience a decline in EDGAR searches relative to the control stocks (G2) after the start of the pilot program. In the regression specification, this would produce a negative coefficient for the interaction term $Post_{i,t} \times Treatment_{i,t}$. Table 5 presents the results of the EDGAR usage analysis. The coefficients for the interaction term $Post_{i,t} \times Treatment_{i,t}$ for the EDGAR *#10-K* and *#10-Q* usage analyses are negative and statistically significant. The EDGAR usage results are thus consistent with dark pool trading increasing information acquisition.

4.2 Type of information being acquired

To further examine information acquisition changes due to dark pool trading, we next evaluate different components of the acquired information. Following Israeli, Lee, and Sridharan (2017) and Glosten, Nallareddy, and Zou (2020), we decompose earnings into an idiosyncratic (firm-specific) component and a systematic component. If the information being acquired is firm specific, we expect an increase in dark pool trading to increase idiosyncratic information sensitivity. If the information being acquired is marketwide, we expect an increase in dark pool trading to increase systematic information sensitivity.

To decompose earnings into its systematic and idiosyncratic components, we perform the following steps. For each stock with earnings data in the past 20 quarters, we estimate the ordinary least squares regression model: $Earnings_{i,t} = a_0 + a_1 Market\ earnings_t + a_2 Industry\ earnings_{i,t} + e_{i,t}$, where $Earnings_{i,t}$ is firm i 's quarterly seasonally adjusted net income before extraordinary items, scaled by the market value of equity at the beginning of quarter t . The variable *Market*

Table 5
Difference-in-differences analysis of EDGAR search volume

Dependent variable =	#10- $K_{i,t}^{-21,-2}$ (1)	#10- $Q_{i,t}^{-21,-2}$ (2)	#other filings $_{i,t}^{-21,-2}$ (3)
$Post_{i,t} \times Treatment_{i,t}$	-0.050** (-2.16)	-0.084* (-1.70)	-10.501 (-0.55)
$Post_{i,t}$	0.040 (1.32)	0.082 (1.10)	-61.661 (-1.06)
$\log(Size)_{i,t}$	-0.060 (-0.90)	-0.320 (-1.14)	2.491 (0.22)
$Book\text{-}to\text{-}market_{i,t}$	0.115 (1.38)	0.005 (0.03)	28.244 (0.94)
$Leverage_{i,t}$	0.043 (0.56)	0.068 (0.24)	-9.720 (-0.20)
$Idiosyncratic\ volatility_{i,t}$	-5.786 (-1.60)	-21.947 (-1.31)	-1,312.997 (-1.24)
$\log(\#Analysts)_{i,t}$	-0.021 (-0.54)	0.045 (0.48)	9.832 (1.03)
$Institutional\ ownership_{i,t}$	0.036 (0.90)	0.059 (0.53)	-11.734 (-0.53)
$Spread_{i,t}$	12.624 (0.95)	70.075 (1.13)	4,677.167 (1.22)
Year-quarter FEs	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes
Observations	1,448	1,448	1,448
Adjusted R -squared	.793	.373	.000

This table reports the estimation results of Equation (6) with the dependent variable taking one of three measures based on EDGAR search volume: #10- $K_{i,t}^{-21,-2}$, #10- $Q_{i,t}^{-21,-2}$, or #Other filings $_{i,t}^{-21,-2}$. Table A.1 in the appendix defines the dependent variables. The variable $Treatment_{i,t}$ equals one for treatment stocks (G3 stocks) and zero for control stocks (G2 stocks). The indicator variable $Post_{i,t}$ equals one for dates after the pilot program is implemented. Standard errors are clustered by firm and earnings announcement date. * $p < .1$; ** $p < .05$; *** $p < .01$.

$earnings_t$ is the market cap-weighted average of $Earnings_{i,t}$ of all firms. The variable $Industry\ earnings_{i,t}$ is the market cap-weighted average of $Earnings_{i,t}$ of firms in the same Fama-French 48 industry as firm i . $Systematic\ earnings_{i,t}$ is computed as the fitted value from the quarterly regression, and $Idiosyncratic\ earnings_{i,t}$ is computed as the residuals of the regression.

We take the decomposed earnings and include both in the Equation (5) specification, replacing $Earnings$ with its systematic component ($Systematic\ earnings$) and its idiosyncratic component ($Idiosyncratic\ earnings$). The specification is:

$$\begin{aligned} Return_{i,t} = & \alpha + \sum_{k=-1}^1 [\beta_k Systematic\ earnings_{i,t+k} \times Post_{i,t} \times Treatment_{i,t} \\ & + \delta_k Idiosyncratic\ earnings_{i,t+k} \times Post_{i,t} \times Treatment_{i,t} \\ & + \zeta_k Sys.\ Earnings_{i,t+k} \times Post_{i,t} + \eta_k Idio.\ earnings_{i,t+k} \times Post_{i,t} \\ & + \kappa_k Sys.\ Earnings_{i,t+k} \times Treatment_{i,t} + \lambda_k Idio.\ earnings_{i,t+k} \end{aligned}$$

$$\begin{aligned}
& \times \text{Treatment}_{i,t} + \mu_k \text{Sys. earnings}_{i,t+k} + \pi_k \text{Idio. earnings}_{i,t+k} \\
& + \Phi' \text{Sys. earnings}_{i,t+k} \times X_{i,t} + \psi' \text{Idio. earnings}_{i,t+k} \times X_{i,t} \\
& + \beta_2 \text{Post}_t \times \text{Treatment}_t + \gamma' X_{i,t} + \theta_i + \theta_t + \varepsilon_{i,t}.
\end{aligned} \tag{7}$$

where the subscripts i and t refer to firm i and year-quarter t , respectively. $\text{Return}_{i,t}$ is defined in Section 1. The indicator variable $\text{Treatment}_{i,t}$ equals one for treatment firms and zero for control firms. The indicator variable $\text{Post}_{i,t}$ equals one for dates after the pilot program is implemented. The vector $X_{i,t}$ is the same set of control variables as in Equation (3). The variables of interest are the interaction terms $\text{Idiosyncratic earnings}_{i,t+1} \times \text{Post}_{i,t} \times \text{Treatment}_{i,t} (\delta_1)$ and $\text{Systematic earnings}_{i,t+1} \times \text{Post}_{i,t} \times \text{Treatment}_{i,t} (\beta_1)$. If dark pool trading facilitates the acquisition of firm-specific information, then the decrease in the level of dark pool trading (resulting from the implementation of the pilot program) should lead to a smaller association between current returns and the future idiosyncratic component of earnings. That is, we expect a negative coefficient for $\text{Idiosyncratic earnings}_{i,t+1} \times \text{Post}_{i,t} \times \text{Treatment}_{i,t} (\delta_1)$. If dark pool trading instead increases the acquisition of systematic information, we expect a negative coefficient for $\text{Systematic earnings}_{i,t+1} \times \text{Post}_{i,t} \times \text{Treatment}_{i,t} (\beta_1)$. Similar to Equation (5), we include firm fixed effects, θ_i , and year-quarter fixed effects, θ_t . We cluster standard errors by firm and year-quarter.

Table 6 presents the estimation results of Equation (7). Column 1 reports results without, and column 2 shows results with, interactions between $\text{Systematic earnings}_{i,t+1}$ and $\text{Idiosyncratic earnings}_{i,t+1}$ and each control variable. In both specifications, the coefficient estimates on the interaction term $\text{Idiosyncratic earnings}_{i,t+1} \times \text{Post}_{i,t} \times \text{Treatment}_{i,t}$ are negative and statically significant. In contrast, the coefficient estimates on the interaction term $\text{Systematic earnings}_{i,t+1} \times \text{Post}_{i,t} \times \text{Treatment}_{i,t}$ are statically insignificant. These results suggest that dark pool trading facilitates the incorporation of firm-specific information into stock prices.

5. Evidence of Informed Trading in Dark Pools

Theory suggests that dark pools facilitate information acquisition by providing a low-cost option for informed traders to trade on their information (Boulatov and George 2013). In this section, we test for evidence to support this proposed mechanism. We expect that the increase in the costs of trading in dark pools following the implementation of the Tick Size Pilot Program will discourage investors from acquiring private information and decrease the amount of informed trading in those pools. To quantify the changes in informed trading in dark pools, we study short selling activity in off-exchange venues as well as three other proxies of informed trading. All the results suggest a

Table 6
Difference-in-differences analysis of effect of dark pool trading on the relation between current return and the components of future earnings

Dependent variable =	<i>Return_{i,t}</i>	
	(1)	(2)
<i>Post_{i,t} × Treatment_{i,t} × Idiosyncratic earnings_{i,t+1}</i>	−2.381** (−2.03)	−3.869** (−2.46)
<i>Post_{i,t} × Treatment_{i,t} × Systematic earnings_{i,t+1}</i>	−0.838 (−0.59)	1.133 (0.46)
<i>Post_{i,t}</i>	−0.011 (−0.22)	−0.031 (−0.59)
<i>Post_{i,t} × Treatment_{i,t}</i>	0.006 (0.33)	0.010 (0.51)
Controls	Yes	Yes
<i>Sys. earnings_{i,t−1}, Idio. earnings_{i,t−1}</i>	Yes	Yes
<i>Sys. earnings_{i,t}, Idio. earnings_{i,t}</i>	Yes	Yes
<i>Sys. earnings_{i,t+1}, Idio. earnings_{i,t+1}</i>	Yes	Yes
<i>Sys. earnings_{i,t−1} × Post_{i,t}, Idio. earnings_{i,t−1} × Post_{i,t}</i>	Yes	Yes
<i>Sys. earnings_{i,t−1} × Treatment_{i,t}, Idio. earnings_{i,t−1} × Treatment_{i,t}</i>	Yes	Yes
<i>Sys. earnings_{i,t−1} × Post_{i,t} × Treatment_{i,t}, Idio. earnings_{i,t−1} × Post_{i,t} × Treatment_{i,t}</i>	Yes	Yes
<i>Sys. earnings_{i,t} × Post_{i,t}, Idio. earnings_{i,t} × Post_{i,t}</i>	Yes	Yes
<i>Sys. earnings_{i,t} × Treatment_{i,t}, Idio. earnings_{i,t} × Treatment_{i,t}</i>	Yes	Yes
<i>Sys. earnings_{i,t} × Post_{i,t} × Treatment_{i,t}, Idio. earnings_{i,t} × Post_{i,t} × Treatment_{i,t}</i>	Yes	Yes
<i>Sys. earnings_{i,t+1} × Post_{i,t}, Idio. earnings_{i,t+1} × Post_{i,t}</i>	Yes	Yes
<i>Sys. earnings_{i,t+1} × Treatment_{i,t}, Idio. earnings_{i,t+1} × Treatment_{i,t}</i>	Yes	Yes
<i>Sys. earnings_{i,t−1} × Controls, Idio. earnings_{i,t−1} × Controls</i>	No	Yes
<i>Sys. earnings_{i,t} × Controls, Idio. earnings_{i,t} × Controls</i>	No	Yes
<i>Sys. earnings_{i,t+1} × Controls, Idio. earnings_{i,t+1} × Controls</i>	No	Yes
Year-quarter FEs	Yes	Yes
Firm FEs	Yes	Yes
Observations	1,448	1,448
Adjusted <i>R</i> -squared	.097	.131

This table provides the estimation results of regressing current quarterly stock returns (*Return_{i,t}*) on the interaction terms (*Post_{i,t} × Treatment_{i,t} × Systematic earnings_{i,t+1}*, *Post_{i,t} × Treatment_{i,t} × Idiosyncratic earnings_{i,t+1}*), where the systematic component of future earnings (*Systematic earnings_{i,t+1}*) is computed as the fitted value from the quarterly regression, and the idiosyncratic component of future earnings (*Idiosyncratic earnings_{i,t+1}*) is computed as the residuals of the regression. Standard errors are clustered by firm and year-quarter. Table A1 in the appendix defines the variables. **p* < .1; ***p* < .05; ****p* < .01.

disproportionate decrease in informed trading through dark pools after the pilot program is introduced.

5.1 Short sales in off-exchange venues

Because short sellers are well-informed about future price moves (Boehmer, Jones, and Zhang 2008; Diether, Lee, and Werner 2009; Karpoff and Lou 2010; Engelberg, Reed, and Ringgenberg 2012), we use short sales as our initial measure of informed trades.

We obtain off-exchange short-sale trade data from the FINRA Trade Reporting Facilities (TRF). As of 2017, short sales comprise 38% of a stock’s off-exchange volume. All off-exchange trades in the U.S. are reported under the TRF consolidated reporting category, so TRF data does not allow us to distinguish trades on lit off-exchange venues (e.g., ECNs) and broker/dealer internalizations from trades executed in dark pools. We therefore

use off-exchange short sales reported by the FINRA TRF as a proxy for dark pool short sales.¹⁷

We measure the level of informed trades in off-exchange venues by the ratio of off-exchange short-sale volume divided by total off-exchange trading volume. Since we focus on information acquisition in the preannouncement period, we average the ratio over the trading days $[-21, -2]$ relative to earnings announcements ($Short\ ratio_{i,t}^{-21,-2}$). We estimate the following model:

$$Short\ ratio_{i,t}^{-21,-2} = \alpha + \beta_1 Post_t \times Treatment_i + \beta_2 Post_t + \gamma' X_{i,t} + \theta_i + \theta_t + \varepsilon_{i,t}. \quad (8)$$

Except for the dependent variable, the specification is the same as Equation (4). If dark pools encourage information acquisition by allowing informed investors to hide their orders, we expect that relative to the control stocks (G2), our treatment stocks (G3) will have a lower ratio of off-exchange short-sale volume divided by off-exchange volume after the implementation of the pilot program. In the regression specification, this will produce a negative coefficient for the interaction term $Post_{i,t} \times Treatment_{i,t}$.

Table 7, panel A, presents the results of the short-sale analysis. In column 1 of Table 7, we find that the coefficient for the interaction term $Post_{i,t} \times Treatment_{i,t}$ is negative and statistically significant, indicating a lower short-sale ratio for treated relative to control firms after the introduction of the Tick Size Pilot Program.

We also split our sample based on the positive and negative earnings surprises. Short sales before negative news are more likely to be informed than short sales before positive news. As a result, we expect the decrease in the level of dark pool trading after the implementation of the pilot program to be associated with a decrease in the short-sale ratio for the subsample of negative earnings surprises, but not for the subsample of positive earnings surprises. In columns 2 and 3, we find a negative and significant coefficient for $Post_{i,t} \times Treatment_{i,t}$ only for firms with negative earnings surprises (t -statistic = -2.22 in column 2; t -statistic = -1.05 in column 3). The chi-square test shows that the effects of dark pools are significantly different across the two subsamples (p -value = $.004$). Overall, these results suggest that dark pool trading increases information acquisition.

To rule out the possibility that the decrease in the level of short selling in dark pools is due to a decrease in uninformed shorting activities, such as trades motivated by hedging and liquidity, we compare the return predictability of short selling in dark pools before and after the pilot program. Following

¹⁷ Because of a lack of data, several prior studies use TRF trade data to proxy for dark pool trading (e.g., Menkveld, Yueshen, and Zhu 2017).

Engelberg, Reed, and Ringgenberg (2012), we estimate the following regression model:

$$\begin{aligned} Return_{i,t}^{-1,18} = & \alpha + \beta_1 Post_t \times Treatment_i \times Short\ Ratio_{i,t}^{-21,-2} + \beta_2 Post_t \\ & \times Treatment_i + \beta_3 Treatment_i \times Short\ Ratio_{i,t}^{-21,-2} + \beta_4 Post_t \\ & \times Short\ Ratio_{i,t}^{-21,-2} + \beta_5 Post_t + \beta_6 Short\ Ratio_{i,t}^{-21,-2} \\ & + \gamma' X_{i,t} + \theta_i + \theta_t + \varepsilon_{i,t}, \end{aligned} \tag{9}$$

where the dependent variable is the 20-days cumulative raw returns ($Return_{i,t}^{-1,18}$). We also perform the same regression with the dependent variable being the 20-days cumulative abnormal returns ($CAR_{i,t}^{-1,18}$). We compute daily abnormal returns as described in Section 1. The variable $Short\ Ratio_{i,t}^{-21,-2}$ is defined the same as in Equation (8). Table 7, panel B, reports the results. The variable of interest is the triple interaction term $Post_t \times Treatment_i \times Short\ Ratio_{i,t}^{-21,-2}$. If the decrease in the level of short selling in dark pools is driven by a decrease in informed short selling, we would expect the return predictability of short selling in dark pools to decrease after the pilot program, that is, a positive coefficient for the triple interaction variable. In contrast, if the short selling reduction in dark pools is driven by a

Table 7
Difference-in-differences analysis of off-exchange short selling

A. Short-sale ratio of off-exchange venues

Dependent variable =	$Short\ ratio_{i,t}^{-21,-2}$		
	Full sample (1)	Negative news (2)	Positive news (3)
$Post_{i,t} \times Treatment_{i,t}$	-0.049*** (-3.49)	-0.115** (-2.22)	-0.021 (-1.05)
$Post_{i,t}$	-0.006 (-0.29)	0.057 (0.76)	-0.026 (-0.92)
$log(Size)_{i,t}$	0.015 (0.59)	-0.002 (-0.02)	-0.011 (-0.31)
$Book-to-market_{i,t}$	0.023 (0.63)	0.192 (1.10)	-0.035 (-0.74)
$Leverage_{i,t}$	0.217** (2.27)	-0.065 (-0.15)	0.144 (0.99)
$Idiosyncratic\ volatility_{i,t}$	1.494** (2.20)	2.241 (1.02)	1.184 (1.26)
$log(\#Analysts)_{i,t}$	0.027 (0.96)	0.046 (0.49)	0.068* (1.67)
$Institutional\ ownership_{i,t}$	0.037 (1.13)	0.041 (0.16)	0.066 (1.24)
$Spread_{i,t}$	2.401* (1.76)	-4.415 (-1.04)	3.300* (1.91)
Chi-square test for equality of coefficients across subsamples	8.50 (0.004)***		
Year-quarter FEs	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes
Observations	1,136	336	800
Adjusted R-squared	.332	.279	.333

(Continued)

Table 7
Continued
B. Return predictability of off-exchange short selling

Dependent variable =	$Return_{i,t}^{-1,18}$	$CAR_{i,t}^{-1,18}$	$Return_{i,t}^{-1,18}$	$CAR_{i,t}^{-1,18}$
	(1)	(2)	(3)	(4)
$Post_{i,t} \times Treatment_{i,t} \times Short\ ratio_{i,t}^{-21,-2}$			0.309*** (2.75)	0.281** (2.52)
$Post_{i,t} \times Treatment_{i,t}$			-0.168*** (-3.75)	-0.148*** (-3.34)
$Treatment \times Short\ ratio_{i,t}^{-21,-2}$			-0.218** (-2.39)	-0.177* (-1.95)
$Post_{i,t} \times Short\ ratio_{i,t}^{-21,-2}$			-0.196** (-2.42)	-0.157* (-1.95)
$Post_{i,t}$			0.121*** (3.39)	0.062* (1.73)
$Short\ ratio_{i,t}^{-21,-2}$	-0.097*** (-3.87)	-0.056** (-2.29)	0.041 (0.69)	0.043 (0.72)
$\log(Size)_{i,t}$			-0.109*** (-2.84)	-0.126*** (-3.24)
$Book\text{-}to\text{-}market_{i,t}$			0.159*** (3.06)	0.140*** (2.61)
$Leverage_{i,t}$			-0.188** (-2.07)	-0.179** (-1.99)
$Idiosyncratic\ volatility_{i,t}$			-1.570* (-1.69)	-1.110 (-1.23)
$\log(\#Analysts)_{i,t}$			0.047 (1.53)	0.053* (1.66)
$Institutional\ ownership_{i,t}$			-0.001 (-0.02)	0.010 (0.17)
$Spread_{i,t}$			0.691 (0.39)	0.337 (0.19)
Year-quarter FEs	No	No	Yes	Yes
Firm FEs	No	No	Yes	Yes
Observations	1,136	1,136	1,136	1,136
Adjusted R-squared	.011	.003	.151	.123

Panel A reports the estimation results of Equation (8), and panel B reports the estimation results of Equation (9). In panel A, the dependent variable is $Short\ ratio_{i,t}^{-21,-2}$ defined as the ratio of off-exchange short-sale volume divided by total off-exchange trading volume over trading days $[-21, -2]$ relative to earnings announcements. In panel B, the dependent variables are the 20-days cumulative raw returns ($Return_{i,t}^{-1,18}$) and the 20-day cumulative abnormal returns ($CAR_{i,t}^{-1,18}$). $Treatment_{i,t}$ equals one for treatment stocks (G3 stocks) and zero for control firms (G2 stocks). The indicator variable $Post_{i,t}$ equals one for dates after the pilot program is implemented. Table A.1 in the appendix defines the variables. In panel A, column 1, performs the analysis on the full DiD sample; column 2 performs the analysis on the DiD earnings announcements that report negative earnings; and column 3 performs the analysis on the DiD earnings announcements that report positive earnings. We report a chi-square test of equality on the $Post_{i,t} \times Treatment_{i,t}$ coefficient between the subsamples in columns 2 and 3. Standard errors are clustered by firm and earnings announcement date. * $p < .1$; ** $p < .05$; *** $p < .01$.

reduction in uninformed short selling, the return predictability of short selling should increase or be unchanged after the pilot program. That is, we expect a negative or insignificant coefficient for the triple interaction variable. X is the same set of control variables that are used in Equation (4). Finally, we include firm fixed effects, θ_i , and year-quarter fixed effects, θ_t , and cluster standard errors by firm and earnings announcement date.

Before discussing the results of the full specification we conduct a simple univariate regression between the short ratio and the future return. In the first two

columns of Table 7, panel B, we find a large and statistically significant negative coefficient. Using the unadjusted return, the coefficient is -0.097 . With the abnormal returns the coefficient is -0.056 . The results suggest that on average, a higher level of short selling in dark pools is associated with lower future returns. Next, we turn to the results from the Equation (9) specification. The coefficient for the variable of interest, $Post_t \times Treatment_i \times Short\ ratio_{i,t}^{-21,-2}$, is positive and statistically significant. The interpretation is that the association between short sales and future stock returns is lower for the treatment stocks after the rule change. The decrease in the association is consistent with short sellers being less informed, which is as expected if information acquisition declines.¹⁸

5.2 Additional indicators of informed trading

The test on short sales is one approach to directly examining the effect of dark pool trading on informed trades. We provide three alternative approaches using the difference-in-differences setting. We consider return synchronicity, the Hasbrouck (1991) permanent price impact, and hedge fund holdings. Each measure is a way to detect informed trading. While the hedge fund holdings and return synchronicity measures only detect informed trading overall, the Hasbrouck permanent price impact can be used to detect off-exchange informed trading.

5.2.1 Return synchronicity. We use return synchronicity as an alternative measure of price informativeness. Following Morck et al. (2000), we measure the return synchronicity, $Synchronicity_{i,t}^{-21,-2}$, as $\ln(R^2/(1-R^2))$. R^2 is estimated using the model $r_{i,t} = r_{i,t}^{Market} + r_{i,t}^{Industry} + \varepsilon_{i,t}$ over trading days $[-21, -2]$ relative to earnings announcements. $r_{i,t}^{Market}$ is market return and $r_{i,t}^{Industry}$ is industry return defined based on Fama-French 48 industry classification. A higher level of *Synchronicity* indicates a lower level of price informativeness. The intuition is that the less firm-specific information is produced and incorporated into the stock price, the better a stock's return is approximated by market and industrywide information (Roll 1988).

$$Synchronicity_{i,t}^{-21,-2} = \alpha + \beta_1 Post_t \times Treatment_i + \beta_2 Post_t + \gamma' X_{i,t} + \theta_i + \theta_t + \varepsilon_{i,t}. \quad (10)$$

Except for the dependent variable, the specification is the same as Equation (4). If dark trading increases price informativeness by allowing informed

¹⁸ In columns 3 and 4 of Table 7, panel B, the coefficient for $Short\ ratio_{i,t}^{-21,-2}$ captures the return predictability of short selling for control firms in preperiod as opposed to the average return predictability. The return predictability of short selling for treated firms in the preperiod is -0.177 ($\beta_6 + \beta_3 = 0.041 - 0.218 = -0.177$) and for treated firm in the post period is -0.064 ($\beta_6 + \beta_4 + \beta_3 + \beta_1 = 0.041 - 0.196 - 0.218 + 0.309 = -0.064$), that is, a 64% decrease in the return predictability from that for treated firms in the preperiod. The DiD effect is economically significant.

investors to hide their orders, we expect that relative to the control stocks (G2), the treatment stocks (G3) will have a higher level of return synchronicity after the implementation of the pilot program.

5.2.2 Hasbrouck permanent price impact. Following Comerton-Forde and Putnins (2015), we use the Hasbrouck (1991) vector auto-regression (VAR) framework to measure the permanent price impact of off-exchange trades. We calculate returns ($r_{i,t}$) and the signed dollar volume of exchange and dark trades (*On-exchange* $vol_{i,t}$, *Off-exchange* $vol_{i,t}$) in each 5-minute interval. For each stock-day we estimate the following system:

$$\begin{aligned} r_{i,t} &= \mu^r + \sum_{\tau=1}^{10} [\lambda_{i,\tau}^r r_{i,t-\tau} + \lambda_{i,\tau}^{On-exchg} On-exchange\ vol_{i,t-\tau} \\ &\quad + \lambda_{i,\tau}^{Off-exchg} Off-exchange\ vol_{i,t-\tau}] + \varepsilon_{i,t}^r \\ On-exchange\ vol_{i,t} &= \mu^{Exchg} + \sum_{\tau=1}^{10} [\phi_{i,\tau}^r r_{i,t-\tau} + \phi_{i,\tau}^{On-exchg} Off-exchange\ vol_{i,t-\tau} \\ &\quad + \phi_{i,\tau}^{Off-exchg} Off-exchange\ vol_{i,t-\tau}] + \varepsilon_{i,t}^{Exchg} \\ Off-exchange\ vol_{i,t} &= \mu^{Off-exchg} + \sum_{\tau=1}^{10} [\theta_{i,\tau}^r r_{i,t-\tau} + \theta_{i,\tau}^{On-exchg} Off-exchange\ vol_{i,t-\tau} \\ &\quad + \theta_{i,\tau}^{Off-exchg} Off-exchange\ vol_{i,t-\tau}] + \varepsilon_{i,t}^{Off-exchg}. \end{aligned} \quad (11)$$

Since we focus on information acquisition in the preannouncement period, we average the permanent price impact of off-exchange trades over the trading days $[-21, -2]$ relative to earnings announcements (*Price impact off-exchg.*^{-21,-2}) and multiply the measure by 100.

$$\begin{aligned} Price\ impact\ off-exchg.^{-21,-2} &= \alpha + \beta_1 Post_t \times Treatment_i + \beta_2 Post_t \\ &\quad + \gamma' X_{i,t} + \theta_i + \theta_t + \varepsilon_{i,t}. \end{aligned} \quad (12)$$

Except for the dependent variable, the specification is the same as Equation (4). If dark pool trading increases the arrival rate of informed trading, we expect that a decrease in the level of dark pool trading after the implementation of the pilot program leads to smaller permanent price impacts of dark trades. That is, we expect to see a negative coefficient for the interaction term $Post_{i,t} \times Treatment_i$ in Equation (12).

5.2.3 Hedge fund holdings. Next, we attempt to detect long-side informed trading by examining hedge fund holding. Hedge fund managers are considered

as well-informed traders. From the long-side perspective, Agarwal et al. (2013) find that the confidential holdings of hedge fund holdings have strong return predictability. Cao et al. (2018) find that undervalued stocks with larger hedge fund ownership realize higher returns. DeVault, Sias, and Starks (2019) show that shocks to hedge fund demand predict stock returns. Choi et al. (2017) show that short sales by hedge funds are highly profitable. If dark pools decrease trading costs for informed traders, hedge funds should become more likely to trade and hold stocks.

We obtain hedge fund holding data from the Thomson Reuters 13-F data. Based on institution names, we classify the 13-F institutions into hedge fund companies and other institutions.¹⁹

$$\begin{aligned} \text{Hedge fund holdings}_{i,t} = & \alpha + \beta_1 \text{Post}_t \times \text{Treatment}_i + \beta_2 \text{Post}_t \\ & + \gamma' X_{i,t} + \theta_i + \theta_t + \varepsilon_{i,t}, \end{aligned} \quad (13)$$

where *Hedge fund holdings*_{*i,t*} is common shares owned by hedge funds divided by total share outstanding at the end of quarter *t*. Except for the dependent variable, the specification is the same as Equation (4). We expect that relative to the control stocks (G2), the treatment stocks (G3) will have a lower level of hedge fund ownership after the implementation of the pilot program. The regression specification will produce a negative coefficient for the interaction term *Post*_{*i,t*} × *Treatment*_{*i,t*}.²⁰

We report the estimation results of Equations (10), (12), and (13) in Table 8. Consistent with our expectations, Table 8 shows a positive and statistically significant coefficient for the interaction term *Post*_{*i,t*} × *Treatment*_{*i,t*} when the dependent variable is *Synchronicity*_{*i,t*}^{-21,-2}, and a negatively significant coefficient when the dependent variables are *Price impact off-exchg.*^{-21,-2} and *Hedge fund holdings*_{*i,t*}.²¹ The results provide additional support that informed trading for treated firms relative to control firms declined after the introduction of the Tick Size Pilot Program. The results in this section are consistent with informed trading decreasing when trading in dark pools becomes more difficult.

¹⁹ We employ the data on hedge fund company names from Cao et al. (2018). Hedge fund companies with more than \$100 million in assets under management (AUM) are required to file quarterly Form 13-F to disclose equity holdings. 13-F filings do not indicate which institutions are hedge fund companies. Cao et al. (2018) compiles a list of hedge fund company names pulled from six commercial hedge fund databases. They matched the list with the names of 13-F institutions to screen for hedge fund companies.

²⁰ Hedge fund holdings are a rough proxy of informed trades. Thomson Reuters 13-F data report institutional holdings at the stock-quarter level. These data do not allow us to directly observe the hedge fund trades or the trading venue handling the hedge fund trades.

²¹ In untabulated results, we find that the coefficient for the interaction term *Post*_{*i,t*} × *Treatment*_{*i,t*} is statistically insignificant for the permanent price impact of exchange trades.

Table 8
Difference-in-differences analysis of informed trading proxies

Dependent variable =	<i>Synchronicity</i> ^{-21,-2} _{<i>i,t</i>}	<i>Price impact</i> <i>off-exchg.</i> ^{-21,-2} _{<i>i,t</i>}	<i>Hedge fund</i> <i>holdings</i> _{<i>i,t</i>}	
	(1)	(2)	(3)	(4)
<i>Post</i> _{<i>i,t</i>} × <i>Treatment</i> _{<i>i,t</i>}	0.243** (2.10)	-0.078*** (-2.64)	-0.006** (-2.31)	-0.006** (-2.14)
<i>Post</i> _{<i>i,t</i>}	0.331 (1.23)	0.023 (0.66)	0.001 (0.21)	0.003 (0.65)
<i>log(Size)</i> _{<i>i,t</i>}	0.303* (1.75)	-0.054 (-1.62)	0.011 (1.51)	0.012* (1.78)
<i>Book-to-market</i> _{<i>i,t</i>}	-0.021 (-0.06)	-0.088 (-0.84)	-0.003 (-0.37)	0.001 (0.10)
<i>Leverage</i> _{<i>i,t</i>}	-0.214 (-0.23)	-0.095 (-0.70)	0.043** (2.00)	0.045* (1.94)
<i>Idiosyncratic volatility</i> _{<i>i,t</i>}	-5.425* (-1.67)	0.368** (2.05)	0.169* (1.72)	0.150 (1.48)
<i>log(#Analysts)</i> _{<i>i,t</i>}	0.297 (1.25)	-0.035 (-1.21)	-0.002 (-0.41)	0.002 (0.29)
<i>Institutional ownership</i> _{<i>i,t</i>}	0.211* (1.83)	-0.006 (-1.07)	0.114*** (5.66)	
<i>Spread</i> _{<i>i,t</i>}	-0.066 (-0.27)	2.707 (1.13)	0.328* (1.68)	0.354* (1.66)
Year-quarter FEs	Yes	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes	Yes
Observations	1,415	1,446	1,459	1,459
Adjusted <i>R</i> -squared	.522	.093	.943	.933

This table reports the estimation results of the Equations (10), (12), and (13) with the dependent variables being return synchronicity (*Synchronicity*^{-21,-2}_{*i,t*}), Hasbrouck's (1991) permanent price impact of off-exchange trades (*Price impact off-exchg.*^{-21,-2}_{*i,t*}), and hedge fund ownership (*Hedge fund holdings*_{*i,t*}). The variable *Treatment*_{*i,t*} equals one for treatment stocks (G3 stocks) and zero for control firms (G2 stocks). The indicator variable *Post*_{*i,t*} equals one for dates after the pilot program is implemented. Table A.1 in the appendix defines the variables. Standard errors are clustered by firm and earnings announcement date in columns 1 and 2, and by firm and year-quarter in columns 3 and 4. **p* < .1; ***p* < .05; ****p* < .01.

6. Alternative Explanations

While we exploit the Tick Size Pilot Program to examine the causal effect of dark pool trading on information acquisition, alternative explanations may describe the findings. Lit venue liquidity, algorithmic trading, and informational efficiency could all conceivably drive the findings, and therefore warrant further investigation. For an alternative explanation to drive the findings, it must be directly affected by the Tick Size Pilot Program (Heath et al. 2019). Therefore, we test whether any of the alternative explanations were directly affected by the pilot program. Across the varying tests, we find no statistically significant evidence that the three alternative explanations drive our findings.

6.1 Lit venue liquidity

The first alternative explanation for our findings is that they are attributable to lit venue liquidity changes. A comparison of effects between G2 and G3 is supposed to hold liquidity constant. However, if the Tick Size Pilot Program affects lit venue liquidity differently for G2 and G3 stocks, then information acquisition for G2 and G3 stocks should differ, and lit venue

liquidity could drive the positive relation between dark pool trading and information acquisition.

To rule out this alternative explanation, we repeat the Equation (4) DiD regression specification with the dependent variable being the Corwin and Schultz (2012) effective spread measure. We average daily Corwin and Schultz’s (2012) effective spread measure over trading days $[-21, 1]$ relative to earnings announcements, which is the same as the construction window of our jump ratio measure.

We present the estimation results in Table 9, panel A. We find an insignificant coefficient for the interaction term $Post_{i,t} \times Treatment_{i,t}$, suggesting that the difference in liquidity between treatment firms and control firms is not significant. Thus, lit venue liquidity does not explain the findings.

6.2 Algorithmic trading

Another possible explanation of our findings is that algorithmic trading (AT) drive them. Weller (2018) and Lee and Watts (2021) show that more AT—including high frequency trading—leads to less information acquisition. If AT is negatively correlated with dark pool trading, then the change in information acquisition that we attribute to dark pool trading could in fact be driven by AT.

To rule out AT as an alternative explanation, we repeat the Equation (4) DiD regression specification with AT proxies as the dependent variables. Since there is no formal, publicly available AT measure, we consider four proxies of AT

Table 9
Difference-in-differences analysis of liquidity and algorithmic trading
A. Difference-in-differences analysis of liquidity

Dependent variable =	$Spread_{i,t}^{-21,1}$
	(1)
$Post_{i,t} \times Treatment_{i,t}$	0.045 (1.24)
$Post_{i,t}$	0.129** (2.21)
$log(Size)_{i,t}$	0.124 (1.34)
$Book-to-market_{i,t}$	0.182 (1.19)
$Leverage_{i,t}$	0.116 (0.42)
$Idiosyncratic\ volatility_{i,t}$	9.319*** (3.28)
$log(\#Analysts)_{i,t}$	0.048 (0.65)
$Institutional\ ownership_{i,t}$	0.066 (0.42)
Year-quarter FEs	Yes
Firm FEs	Yes
Observations	1,465
Adjusted R-squared	.586

(Continued)

Table 9
Continued

B. Difference-in-differences analysis of algorithmic trading

Dependent variable =	<i>Odd-lot ratio</i> _{<i>i,t</i>} ^{-21,1}	<i>Trade-to-order ratio</i> _{<i>i,t</i>} ^{-21,1}	<i>Cancel-to-trade ratio</i> _{<i>i,t</i>} ^{-21,1}	<i>Average trade size</i> _{<i>i,t</i>} ^{-21,1}
	(1)	(2)	(3)	(4)
<i>Post</i> _{<i>i,t</i>} × <i>Treatment</i> _{<i>i,t</i>}	-0.001 (-0.09)	0.002 (1.64)	-0.134 (-0.05)	-0.003 (-1.39)
<i>Post</i> _{<i>i,t</i>}	-0.020** (-2.35)	0.002 (1.13)	-3.313 (-0.89)	0.011*** (3.21)
<i>log(Size)</i> _{<i>i,t</i>}	0.027** (2.49)	-0.003 (-1.17)	-11.294 (-1.31)	-0.030*** (-5.48)
<i>Book-to-market</i> _{<i>i,t</i>}	0.027 (1.53)	-0.003 (-0.76)	4.981 (0.46)	-0.003 (-0.40)
<i>Leverage</i> _{<i>i,t</i>}	-0.030 (-0.96)	0.013 (1.55)	1.950 (0.15)	0.018 (1.33)
<i>Idiosyncratic volatility</i> _{<i>i,t</i>}	-0.518** (-2.12)	0.099* (1.74)	-142.069 (-1.27)	0.033 (0.26)
<i>log(#Analysts)</i> _{<i>i,t</i>}	0.008 (0.73)	-0.001 (-0.42)	1.356 (0.33)	-0.009* (-1.91)
<i>Institutional ownership</i> _{<i>i,t</i>}	0.007 (0.35)	-0.002 (-0.44)	11.306 (1.46)	-0.007 (-1.08)
<i>Spread</i> _{<i>i,t</i>}	-1.606* (-1.91)	0.084 (0.57)	-108.867 (-0.24)	0.363 (1.03)
Year-quarter FEs	Yes	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes	Yes
Observations	1,465	1,465	1,465	1,465
Adjusted <i>R</i> -squared	.744	.646	.538	.756

Panel A reports the estimation results of the Equation (4) DiD regression specification with the dependent variable being *Spread*_{*i,t*}^{-21,1} the Corwin and Schultz (2012) effective spread measure, averaged over trading days [-21, 1] relative to earnings announcements. Note that *Spread*_{*i,t*}^{-21,1} is multiplied by 100 so as to be in percentage units. Panel B reports the estimation results of the Equation (4) DiD regression specification with the dependent variable taking one of four algorithmic trading measures: *Odd-lot ratio*_{*i,t*}^{-21,1}, *Trade-to-order ratio*_{*i,t*}^{-21,1}, *Cancel-to-trade ratio*_{*i,t*}^{-21,1}, or *Average trade size*_{*i,t*}^{-21,1}. *Odd-lot ratio*_{*i,t*}^{-21,1} is the ratio of volume of trades executed in odd-lot sizes divided by the total volume traded; *Trade-to-order ratio*_{*i,t*}^{-21,1} is the ratio of total volume traded divided by the total volume of orders placed; *Cancel-to-trade ratio*_{*i,t*}^{-21,1} is the ratio of number of orders canceled divided by the number of trades executed; and *Average trade size*_{*i,t*}^{-21,1} is the ratio of total volume traded divided by the number of trades. We average the measures of algorithmic trading over trading days [-21, 1] relative to earnings announcements. The variable *Treatment*_{*i,t*} equals one for treatment stocks (G3 stocks) and zero for control firms (G2 stocks). The indicator variable *Post*_{*i,t*} equals one for dates after the pilot program is implemented. Table A.1 in the appendix defines the variables. Standard errors are clustered by firm and earnings announcement date. **p* < .1; ***p* < .05; ****p* < .01.

from prior studies: the odd-lot ratio, the trade-to-order ratio, the cancel-to-trade ratio, and the average trade size.

We obtain the AT proxies from the SEC’s Market Information Data Analytics System (MIDAS) data set. We calculate *Odd-lot ratio*_{*i,t*} as the volume of trades executed in odd-lot sizes divided by the total volume traded. *Odd-lot ratio*_{*i,t*} is positively correlated with AT (O’Hara, Yao, and Ye 2014). *Trade-to-order ratio*_{*i,t*} is the total volume traded divided by the total volume of orders placed. A greater *Trade-to-order ratio*_{*i,t*} is associated with lower AT because algorithmic traders (ATs) place and cancel high numbers of orders

when executing trades (Hendershott, Jones, and Menkveld 2011). *Cancel-to-trade ratio* $_{i,t}$ is the number of orders canceled divided by the number of trades executed. *Cancel-to-trade ratio* $_{i,t}$ is positively correlated with AT, as ATs place and cancel large numbers of orders (Hasbrouck and Saar 2013; Hendershott and Riordan 2013). The final proxy, *Average Trade Size* $_{i,t}$, is the total volume traded divided by the number of trades. The level of AT is expected to be negatively correlated with the *Average trade size*, as ATs execute a greater number of small orders to trade a given volume (Brogaard, Hendershott, and Riordan 2014; Menkveld 2014; O'Hara 2015). We average daily AT measures over trading days $[-21, 1]$ relative to earnings announcements, which is the same as the construction window of our jump ratio measure.

We repeat the Equation (4) DiD regression specification with each measure of AT as the dependent variable. Table 9, panel B, presents the results of DiD analysis. In columns 1–4, none of the coefficients for the interaction term $Post_{i,t} \times Treatment_{i,t}$ is statistically significant, suggesting that the difference in AT between treatment firms and control firms is not significant.²² The results do not support the alternative explanation that AT drives the results.²³ It is worth noting that the coefficient for $Post_{i,t}$ is negative and statistically significant, consistent with the findings of Lee and Watts (2021) that G1, G2, and G3 overall experienced a decline in algorithmic trading, resulting in greater information acquisition.

6.3 Informational efficiency

The third alternative explanation we explore is that the Tick Size Pilot Program changes the informational efficiency of the market. Zhu (2014) posits that dark pools attract uninformed traders, while informed traders gravitate to the lit markets. This bifurcation could lead to clearer signals of supply and demand in the lit markets and thereby improve informational efficiency. Consistent with Zhu (2014), Comerton-Forde and Putninš (2015) find that a rise in dark pools leads to an increase in informational efficiency. A change in the market's informational efficiency may lead to a change in information acquisition (Weller 2018). Thus, informational efficiency could drive the relation we observe between dark pool trading and information acquisition.

To rule out this alternative explanation, we consider three measures of informational efficiency: variance ratio, the Hou and Moskowitz (2005) stock price delay measure, and autocorrelation. The variance ratio, $Abs(Variance\ ratio)^{-21,1}$, is defined as the absolute value of one minus the ratio of the

²² The results are qualitatively unchanged when we use the log transformation of the AT variables.

²³ In untabulated results we also repeat the main analysis, Tables 2 to 4, including the AT variables as control variables and find no qualitative impact on the results. Lee and Watts (2021) use the same Tick Size Pilot Program to show algorithmic trading decreases information acquisition. Their results are not inconsistent with ours as they use the three affected groups of stocks (G1, G2, and G3) as their treatment, whereas we narrow in on G3 versus G2 (treatment versus control) difference.

variance of 2-day returns divided by two times the variance of 1-day returns. The variances are computed over trading days $[-21, 1]$ relative to earnings announcements. If the variance ratio is closer to zero, prices behave more like a random walk and the stock price is more efficient. Hou and Moskowitz (2005) use stock prices' delay in reflecting marketwide information to measure informational efficiency. Price delay, $Price\ delay^{-21,1}$, is measured as one minus the ratio of a stock's daily return variation explained by the concurrent daily market return divided by a stock's daily return variation explained by the concurrent and four past daily market returns. The regressions are estimated over trading days $[-21, 1]$ relative to earnings announcements. If the measure is lower (approaching zero), less stock return variation is captured by lagged daily market returns, and the stock price is more efficient. Lastly, we measure the first order return autocorrelations, $Autocorrelation^{-21,1}$, over trading days $[-21, 1]$ relative to earnings announcements. Because new information affecting returns should be unpredictable, returns should not be predicted by past information, such as past returns. Therefore, if $Autocorrelation^{-21,1}$ is closer to zero, the stock price is more efficient. We construct the informational efficiency measures over the same window as that of our jump ratio measure.

To test whether informational efficiency changed, we repeat the Equation (4) DiD regression specification with the dependent variable taking one of the three informational efficiency measures. The variable of interest is therefore the interaction term $Post_{i,t} \times Treatment_{i,t}$ in each specification.

Table 10 reports the results. For the three dependent variables, the coefficients for the interaction term $Post_{i,t} \times Treatment_{i,t}$ are statically insignificant, suggesting that the difference in the informational efficiency between treatment firms and control firms is not significant. Note that in two of the three specifications we find none of the independent variables to be statistically significantly different from zero. We are not surprised by the finding that information efficiency does not change for the treatment stocks relative to the control stocks following an earnings announcement. The theoretical predictions for the relation between price efficiency and dark pool trading vary. It could be benign, beneficial, or detrimental. For example, as dark pools divert a large volume of trades from exchanges, the market depth on exchanges decreases, transitory price movements increase, and thus leading to less informational efficiency (Hatheway, Kwan, and Zhen 2017). Alternatively, if the introduction of dark pools results in a partial separating equilibrium whereby uninformed traders are more likely to use dark pools and informed traders are more likely to use lit markets, then informational efficiency may increase (Zhu 2014; Comerton-Forde and Putniņš 2015). We conclude that the information acquisition results are not driven by changes in informational efficiency.

7. Conclusion

We examine the effect of dark pool trading on the incorporation of information into stock prices. Given dark pools' opacity and rapidly growing market share,

Table 10
Difference-in-differences analysis of informational efficiency

Dependent variable =	$Pricedelay_{i,t}^{-21,1}$	$Abs(Varianceratio)_{i,t}^{-21,1}$	$Autocorrelation_{i,t}^{-21,1}$
	(1)	(2)	(3)
$Post_{i,t} \times Treatment_{i,t}$	0.064 (1.37)	-0.054 (-0.59)	0.009 (0.35)
$Post_{i,t}$	0.022 (0.32)	-0.002 (-0.01)	0.044 (1.08)
$log(Size)_{i,t}$	-0.043 (-0.45)	-0.438 (-1.55)	-0.056 (-0.74)
$Book-to-market_{i,t}$	-0.072 (-0.51)	0.005 (0.01)	0.011 (0.11)
$Leverage_{i,t}$	-0.026 (-0.08)	-0.064 (-0.17)	0.002 (0.01)
$Idiosyncratic\ volatility_{i,t}$	11.709*** (4.64)	2.066 (0.31)	-2.053 (-1.50)
$log(\#Analysts)_{i,t}$	-0.035 (-0.34)	-0.110 (-0.55)	-0.072 (-1.33)
$Institutional\ ownership_{i,t}$	0.085 (0.39)	-0.120 (-0.49)	-0.001 (-0.01)
$Spread_{i,t}$	-9.745** (-2.00)	4.314 (0.43)	-2.474 (-0.85)
Year-quarter FEs	Yes	Yes	Yes
Firm FEs	Yes	Yes	Yes
Observations	1,465	1,465	1,465
Adjusted R-squared	.411	.003	.040

This table reports the estimation results of the Equation (4) DiD regression specification with the dependent variable taking one of three measures of informational efficiency: $Price\ delay_{i,t}^{-21,1}$, $Abs(Varianceratio)_{i,t}^{-21,1}$, or $Autocorrelation_{i,t}^{-21,1}$. The variable $Treatment_{i,t}$ equals one for treatment stocks (G3 stocks) and zero for control stocks (G2 stocks). The indicator variable $Post_{i,t}$ equals one for dates after the pilot program is implemented. Table A.1 in the appendix defines the variables. Standard errors are clustered by firm and earnings announcement date. * $p < .1$; ** $p < .05$; *** $p < .01$.

regulatory authorities are increasingly concerned about the impact of dark pool trading on price discovery. Our central conjecture is that dark pool trading can influence price discovery through its influence on information acquisition.

We find that more dark pool trading is linked with greater information acquisition. Using newly available weekly dark pool trading data for a comprehensive sample of firms, we find that dark pool trading is associated with a greater preemption of earnings announcements and a stronger relation between current returns and future earnings. Using the differential treatment of the Tick Size Pilot Program as an exogenous shock to the level of dark pool trading, we find that the positive relationship between dark pool trading and information acquisition is causal.

We examine a number of possible alternative explanations and find that none explain the findings. In addition, through an analysis of SEC EDGAR searches, we show direct evidence that dark pool trading increases information acquisition. And when we bifurcate earnings into a firm-specific component and a systematic component, we show that dark pool trading leads to greater information acquisition for the firm-specific component. Overall, the evidence is consistent with dark pool trading increasing information acquisition.

Appendix

Table A.1
Variable definitions

Variable	Definition
Dark pool trading variables	
<i>Dark ratio</i>	Ratio of the trading volume executed in dark pools divided by the consolidated volume over the preannouncement period. The preannouncement period includes all weeks between the earnings announcement date of quarter $t-1$ and 21 trading days before the announcement date of quarter t
<i>Dark ratio qtr</i>	Ratio of the trading volume executed in dark pools divided by the consolidated volume in month $t-2$ to month t relative to the fiscal quarter end
Other variables	
<i>Jump</i> ^{-21,1}	The ratio of cumulative abnormal return over trading days $[-1, 1]$ around earnings announcements divided by cumulative abnormal return over trading days $[-21, 1]$ around earnings announcements: $CAR^{-1,1}/CAR^{-21,1}$. Daily abnormal returns are computed as the raw return less the buy-and-hold return to a benchmark portfolio of firms matched on size and the book-to-market ratio. The benchmark portfolios are constructed using Fama and French's (1992) method. All firms with CRSP share codes 10 and 11 are classified into 25 portfolios by size at the end of June of year t and by the book-to-market ratio at the end of December of year $t-1$
<i>Return</i>	The buy-and-hold return over month $t-2$ to month t relative to the fiscal quarter end
<i>Earnings</i>	Seasonally adjusted net income before extraordinary items scaled by the market value of stock i at the beginning of the quarter
<i>Size</i>	Market value of common shares
<i>Book-to-market</i>	Book value of common shares divided by the market value of equity
<i>Leverage</i>	Total debt (Compustat data item DLTT plus data item DLC) divided by total assets (Compustat data item AT)
<i>Growth</i>	Growth in assets
<i>Idiosyncratic volatility</i>	The variance of the residual obtained by fitting the Carhart (1997) four-factor model to the time series of daily stock returns
<i>#Analysts</i>	Number of analysts following the firm
<i>Institutional ownership</i>	Institutional ownership
<i>Loss</i>	Indicator variable taking the value of one if income before extraordinary items (Compustat item IB) in quarter $t+1$ is negative and zero otherwise
<i>Spread</i>	Corwin and Schultz (2012) effective spread measure, averaged over quarter t
<i>Spread</i> ^{-21,1}	Corwin and Schultz (2012) effective spread measure, averaged over trading days $[-21, 1]$ relative to earnings announcements
<i>Odd-lot ratio</i> ^{-21,1}	The ratio of volume of trades executed in odd-lot sizes divided by the total volume traded, averaged over trading days $[-21, 1]$ relative to earnings announcements
<i>Trade-to-order ratio</i> ^{-21,1}	The ratio of total volume traded divided by the total volume of orders placed, averaged over trading days $[-21, 1]$ relative to earnings announcements
<i>Cancel-to-trade ratio</i> ^{-21,1}	The ratio of number of orders canceled divided by the number of trades executed, averaged over trading days $[-21, 1]$ relative to earnings announcements
<i>Average trade size</i> ^{-21,1}	The ratio of total volume traded divided by the number of trades, averaged over trading days $[-21, 1]$ relative to earnings announcements

(Continued)

Table A.1
Continued

Variable	Definition
$Price\ delay^{-21,1}$	One minus the ratio of a stock's daily return variation explained by the concurrent daily market return divided by a stock's daily return variation explained by the concurrent and four past daily market returns. The regressions are estimated over trading days $[-21, 1]$ relative to earnings announcements
$Abs(variance\ ratio)^{-21,1}$	The absolute value of one minus the ratio of the variance of 2-day returns divided by two times the variance of 1-day returns. The variances are computed over trading days $[-21, 1]$ relative to earnings announcements
$Autocorrelation^{-21,1}$	First order return autocorrelations. The autocorrelations are computed over trading days $[-21, 1]$ relative to earnings announcements
$\#10\text{-}K^{-21,-2}$	Average number of downloads for 10-K filings over trading days $[-21, -2]$ relative to earnings announcements, scaled by trading volume, and finally multiplied by 100
$\#10\text{-}Q^{-21,-2}$	Average number of downloads for 10-Q filings over trading days $[-21, -2]$ relative to earnings announcements, scaled by trading volume, and finally multiplied by 100
$\#Other\ filings^{-21,-2}$	Average number of downloads for all other EDGAR filings over trading days $[-21, -2]$ relative to earnings announcements, scaled by trading volume, and finally multiplied by 100
$Return^{-1,18}$	Raw return accumulated over trading days $[-1, 18]$ around earnings announcements
$CAR^{-1,18}$	Cumulative abnormal return over trading days $[-1, 18]$ around earnings announcements. Daily abnormal returns are computed as the raw return less the buy-and-hold return to a benchmark portfolio of firms matched on size and the book-to-market ratio. The benchmark portfolios are constructed using Fama and French's (1992) method. All firms with CRSP share codes 10 and 11 are classified into 25 portfolios by size at the end of June of year t and by the book-to-market ratio at the end of December of year $t - 1$
$Hedge\ fund\ holdings$	Common shares owned by hedge funds divided by total share outstanding at the end of quarter t
$Price\ impact\ off\text{-}exchg.^{-21,-2}$	Permanent price impact of off-exchange trades estimated by the Hasbrouck (1991) vector auto-regression framework at stock-day level. We average the estimated daily permanent price impact over trading days $[-21, -2]$ relative to earnings announcements and multiplied by 100
$Synchronicity^{-21,-2}$	$\ln(R^2/(1-R^2))$, where R^2 is estimated using the model $r_{i,t} = r_{i,t}^{Mkt} + r_{i,t}^{Ind} + \varepsilon_{i,t}$, where $r_{i,t}^{Mkt}$ is market return and $r_{i,t}^{Ind}$ is industry return defined based on Fama-French 48 industry classification. The model is estimated over trading days $[-21, -2]$ relative to earnings announcements
$Short\ ratio^{-21,-2}$	Ratio of off-exchange short-sale volume divided by total off-exchange trading volume over trading days $[-21, -2]$ relative to earnings announcements

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